

Banking Regulation

Foundation of Risk Management (APM_5FI11_AE) – Part I

Disclaimer: This document is made for educational purpose for a lecture given at the ENSAE and must exclusively be used in this context. The author cannot be held liable for any use outside of this context. The content, positions, opinions exposed in this document and throughout the lecture are solely incumbent to the author of this document and do not necessarily reflect whenever applicable, the views of Deloitte.

Foreword: This document complements the slides presented during the lecture and repeats some of the elements presented in the slides. In principle, the slides should be self-sufficient to take the final exam, but we highly recommend the interested reader considering a thorough reading of the present document. All potential errors remaining in this document are incumbent to the author who would be grateful to receive any feedback / correction / suggestion to amend the document accordingly.

Table of Contents

1.	Introduction to banking mechanisms	3
1.1	Reminder of basic accounting principles & representation	4
1.2	Overview of the main risks underpinning the banking business	7
1.3	Summary: role of the Capital	18
1.4	How to cover the liquidity risk(s)?	19
2.	Overview of the key features related to CRR (EU Basel agreement translation).....	20
3.1	The 3 Pillars.....	21
3.2	Main features from Pillar 1 - Capital	22
3.2.1	Introduction to the Solvency Ratio	22
3.2.2	Focus on certain RWA components – Credit Risk	23
3.2.3	Focus on certain RWA components – Operational Risk.....	25
3.2.4	Focus on certain RWA components – Market Risk	25
3.2.5	Focus on certain RWA components – Counterparty Credit Risk.....	26
3.2.6	Focus on the regulatory capital	27
3.2.7	Leverage Ratio	28
3.3	Main features from Pillar 1 – Liquidity / Funding	29
3.3.1	Constraints on Liquidity – Net Stable Funding Ratio	29
3.3.2	Constraints on Liquidity – Liquidity Coverage Ratio	30
3.4	What is the Pillar 2	31
3.4.1	Pillar 2 on Capital – ICAAP	32
3.4.2	Pillar 2 on Liquidity – ILAAP.....	34
3.5	How do supervisors ensure that the Pillar 1 are ensured in practice?	35
3.5	Beyond Pillar 1 & Pillar 2: recovery and resolution framework	35
3.6	Banking regulation – the main actors	35
3.6	.1Basel Committee	36
3.6.2	Banking Regulation & Supervision in the EU (European Union)	36
3.6	Banking Supervision & Supervision in the US	37
4	Conclusion.....	38

5	Appendix	38
5.1	Alternative approach to measure Op Risk on Pillar 1	Error! Bookmark not defined.
5.2	Additional details on Market Risk (standard approaches) on Pillar 1	38
5.3	Additional details to measure Counterparty Credit Risk (“CCR”) on Pillar 1	39
5.4	From Basel Agreements to EBA	40
6	Bibliography	41
6.1	Textbooks	41
6.2	Some key regulatory texts	41
6.2.1	EU regulation on solvency & liquidity	41
6.2.2	EU Guidelines related to ICAAP & ILAAP	42
6.3	Websites	42
7	Index	42

1. Introduction to banking mechanisms

After all, what is banking business all about? It is usually a private undertaking (unless the shareholder is the State) aiming at making profit by transforming “money into money”! In essence, banking business acts as an **intermediary** between those clients which have an excess of savings and those which are seeking for a financing (e.g. mortgage loan, consumer loan). Banks have also extended their activities beyond their original business, denoted as **Retail Banking** or **Commercial Banking** and developed the **Corporate and Investment Banking “CIB” activity also called Wholesale Banking** (e.g. securities & derivatives trading, managing client funds, Merger & Acquisition Operations ...). This lecture will mostly address the risks & regulation stemming from the original banking business (i.e. taking on client deposits and granting loans) and to a certain extent the CIB / wholesale banking.

However, why would a natural person (i.e. a physical person) or a moral person (e.g. a Corporate, a Small and Medium Enterprise “SME”, a Retail Client) with an excess of cash become the client of a bank, i.e. accept to make a deposit of their excess cash into the bank, knowing that the bank will use their funds to grant a loan to other clients?

Prior to answering this question, let us make a parallel with the visit to a dentist (or a doctor). Why would anyone be willing to be examined (or have their teeth drilled) by an individual they laid the eyes upon five minutes before entering into his cabinet? Or to put it differently, what triggers this level of confidence to an unknown individual and by extension to an “unknown” corporation where one deposits its excess cash?

Medicine business and banking business are both strongly regulated: doctors need to have a specific diploma obtained after achieving long studies. This diploma aims at building immediate trust from their patients. So is the banking business, whereby a bank needs to first obtain a **regulatory agreement** to operate (e.g. take on client deposits); this agreement is granted by **supervisory bodies** (e.g. European Central Bank – “ECB”, Autorité de Contrôle Prudentiel et de Résolution “ACPR” in France) after a lengthy (and costly) process in which the bank demonstrates its compliance with a set of standards that we will expose in the remainder of this lecture. Besides, banks remain under ongoing monitoring from the supervisory bodies who receive on a regular basis so-called **“regulatory reportings”** ensuring the enforcement of the applicable regulatory requirements on a continuous basis; in parallel supervisory bodies also conduct **“on-site” inspections** to solely not rely upon regulatory reportings but also to ensure how the bank is managed in practice and also challenge the bank’s senior management.

The remainder of this lecture is organized as follows: i) we will first present the main risks applicable to the banking business through a series of practical cases and introduce the role of the **capital and liquidity** to cover these risks, ii) we will then present how banking regulation and supervision are organized (with a focus on the European Union), iii) we then finish by introducing the main features related to the current banking regulation in force (**Basel III and with a glimpse into Basel IV**), again with a focus in the European Union (**Capital Requirements Regulation & Directive – “CRR” and “CRD”**) as it represents a sufficiently mature market where the banking regulation standards are extremely stringent.

1.1 Reminder of basic accounting principles & representation

The common language to represent, compute and measure the performance and the risks of any banking business is ... the accounting standards. Overlooking the accounting principles is not an option for anyone willing to work in the banking environment (barring exception traders’ bonuses are determined upon their accounting profits; all banking regulations refer as a starting point to accounting measurements). The objective of this section is not to turn you into accounting experts, but solely to make you familiar with the basic accounting mechanisms, nonetheless profound, which serve as our common language. First and foremost, the accounting is grounded on the **“double entry principle”**¹ which requires to account each entry into two separate accounts to materialize both the origin and the destination of the operation. For instance, when a bank’s client makes a deposit of 100, the bank accounts for a liability entry of 100 labelled as “deposit” (as the bank still owes the money to its client – origin of the funds) and an asset entry of 100 labelled as “cash” to materialize that the bank holds the cash.

Although there exist different **generally accepted accounting principles** (“Gaap”) across different jurisdictions, we will adopt the IFRS (International Financial Reporting Standards) which enable to present in a convenient way both retail and wholesale (CIB) businesses by using notions you have already met in practice (e.g., marked-to-market or marked-to-model for the derivatives).

Without further ado, let us expose the paradigmatic **balance sheet** for: i) a retail bank, ii) a CIB. Recall that the balance sheet represents the possession (assets) and the resources (liabilities) available at a given point in time t .

Assets	Liabilities
Loans (Retail, Corporates, Institutional, Interbank)	Sight Deposits Term Deposits Retail (natural person) clients, Small and Medium Enterprises (SMEs), Large Corporate, Financial Institutions
Client Overdrafts	Regulated Savings (e.g. Livret A, PEL/CEL, other)
Cash (unused) and Cash Equivalents	Equity (shareholders)

Figure 1 – Typical retail banking balance sheet

Let us make the following remarks related to the balance sheet of a retail bank:

- **Sight deposits** correspond to amounts in cash deposited by the clients which can be withdrawn at **any time** (and without notice period). For instance, your personal bank account is registered as a liability entry from the bank’s viewpoint as it represents a resource which can be used by the bank to grant loans.

¹ Invented by an Italian mathematician Luca Pacioli in the 15th Century ([Luca Pacioli - Wikipedia](#))

- **Term deposits** are in essence similar to sight deposits but cannot be withdrawn before the contractual term of the deposits.
- **Overdrafts** correspond to temporarily loans granted to clients whose personal bank account becomes negative (“**autorisation de découvert**”).
- All banks are required to hold a bare amount of **cash** which cannot be lent to any client to minimize the **liquidity risk**; this minimal amount is placed at the Central Bank (e.g. the French Central Bank) as “**mandatory reserve**”. Recent regulations imposed additional constraints on liquidity, on top of the mandatory reserves (we will come back to this notion of liquidity risk thereunder).
- The figure #1 is **at scale**, and henceforth clearly illustrates that loans are to a very large extent financed by client deposits and to a small extent by the Equity (also called capital) brought by the shareholders. It becomes obvious that in case a bank would go bankrupt and be liquidated, clients who have a deposit in the bank may be losing part / totality of their savings and would create strong turmoil. Thereby, retail banking activity should be strongly supervised to avoid such a situation.

Let us now present the **paradigmatic** balance sheet a wholesale bank (CIB).

Asset	Liabilities
A: Positive value of the hedging portfolios	B: Negative Market Value (« Fair Value ») of derivative portfolios (C_T) Swaps, options ... Notional values of the derivatives are accounted off-balance sheet
C: Positive Market Value (« Fair Value ») of derivative portfolios Swaps, options ... Notional values of the derivatives are accounted off-balance sheet	D: Negative market value of the derivative hedging portfolios
Securities trading portfolios	Bond Issuances EMTN, Convertibles,...
Cash	Equity (shareholders)

Figure 2 –Typical wholesale banking (CIB) balance sheet

Let us make the following remarks related to the wholesale bank balance sheet:

- The **market value** of the derivatives (also called Fair Value) corresponds to the value stemming from the models (and calibrated onto market parameters) you may have already studied in your previous tenure. You may observe that the derivative’s value generates accounting entries both on the asset and the liability side.
- If you consider that a bank is selling an option (e.g., a short position in a vanilla call, $-(S_t - K)^+$), the payoff is never positive hence generating a liability from the bank perspective, whilst its hedging portfolio is represented on the asset side. The opposite scheme applies when a bank purchases an option.
- If you consider a bank trading an **interest rate swap**, you may remember that the market value of the swap, let us assume a receiver-swap (i.e., the bank receives the fixed rate and pays the floating rate), is equal to the difference between the (present) value of the fixed-rate leg cash-flows minus those from the floating-rate leg. The sign (positive or negative) of the difference between those two legs is unknown and may change in accordance with market conditions. As such, the market-value of an interest rate swap (and by extension to any non-optional derivative, such as Futures, Forward) may be either accounted on the asset side (when the market value is positive from the bank’s viewpoint) or the liability side (when the market value is negative from the bank’s viewpoint).

- Besides the derivatives activities, a wholesale bank may also act as a **market-maker on bonds, stocks** and henceforth hold on its balance sheet a large inventory of bonds, stocks to be traded and meet client demands.

Traditional banking businesses operate under the regulatory license called **“credit institution”** (which entails the maximal regulation and supervision) and under this license may be authorized to perform CIB activities. In other words, from a strict regulation viewpoint, the word banking is a misnomer; we speak of credit institution. Barring exception, we will use without distinction either the word **bank** or credit institution (or simply institution) in this document. It is still possible to solely operate as wholesale banking (i.e. without being entitled to taking on client deposits and granting loans) under a specific regulatory license denoted as **“investment firm”** license, which, depending on the size & complexity of the activities run by the investment firm, may impose less stringent (or equal) regulation and supervision than that of a credit institution.

Banks running both activities (retail and wholesale)² usually display sizeable balance sheets (over EUR 1,000 bn) and denoted by the regulators as “systemic-bank” as their bankruptcy could lead to material damages to the economy and the financial markets. Needless to say, that systemic banks are subject to even more stringent regulation and additional scrutiny from the supervisory bodies.

Back in the years (e.g. Glass Stegall Act- “GSA” in 1933 under the Roosevelt administration in the US, French Banking system in the aftermath of the second world war) wholesale (CIB) banks were neither entitled to take on deposits nor grant loans whilst retail banks were not entitled to engage into wholesale activities such as derivatives or securities trading, structured finance and could not be a subsidiary of a retail bank, keeping hence a “watertight” door between these two businesses.

Besides, wholesale business could only rely upon its **private shareholders** to finance itself, namely through the non-distributed profits accumulated over the years. Recall also, that in the US and in some other countries the shares of the wholesale banks could not be listed on the stock markets, hence structurally limiting their size and the risk they were facing. These separation laws have been gradually dismantled (e.g., GSA was terminated in 1999 under the Clinton Administration, the Debré-Haberer reform in 1966 followed by the banking reform in 1984 put an end to the French banking system established in 1945) and enabled retail banking to engage into wholesale banking activities and conversely.

It is worth noting that recent attempts to resume banking separation (e.g., French Banking Law, Volcker Rule in the US in 2014-2015) have mostly led to streamline securities & derivatives trading activities but did not impose a split between two independent legal entities (as in was under the GSA for instance), one being in charge of retail and the other of wholesale banking. In other words, retail client deposits can still be used to finance trading business. We will not develop the banking separation any further.

In the EU, a **significant** bank displays a balance sheet greater the EUR 30 billion³; a “small” bank displays a balance sheet smaller than EUR 5 to 8 billion; for instance, the balance sheet of BNP Paribas or Bank of America exceeds EUR 2,000 billion.

Let us finally expose in a purportedly simplified version the main building blocks of the profit and loss **“P&L”** which measure **over a year** the gains (or the loss) incurred by the bank.

² This business model is extremely popular in France, where 4 French Banks are universal and systemic.

³ This threshold is a necessary condition to be eligible to the status of systemic bank.

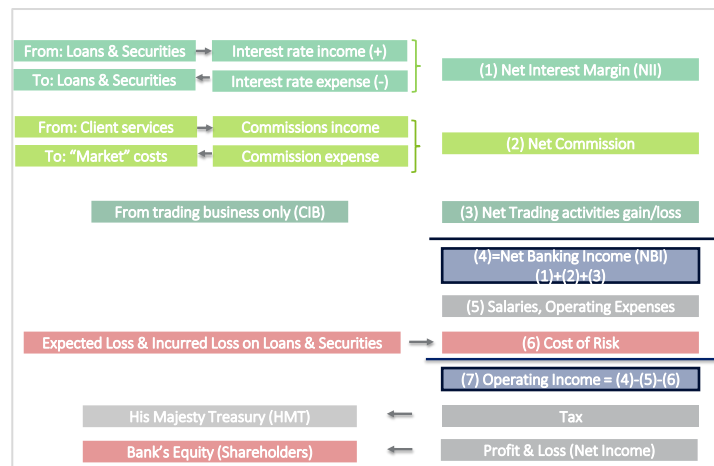


Figure 3 –Typical banking P&L structure

In contrast to the Corporate Business, a bank does not have any turnover (i.e., the sum of the annual sales), but a Net Banking Income ("NBI") which is the (algebraic) sum of (see figure #3):

- the Net Interest Margin ("NII"): represents the difference (positive or negative) between the revenues from the loans granted to the clients less the cost of the resources (e.g., client deposits, bond issuances).
- the Commissions: represents the difference (positive or negative) between the fees charged to clients (e.g., overdrafts – "découverts", securities and trading commissions) and those incurred by the bank.
- the trading revenues represents the gain / loss stemming from the securities & derivatives trading activities.
- the **Cost of Risk "CoR"** (always negative) corresponds to the statistical (potential) and/or incurred loss on the client loans (i.e., loans which are likely or highly likely to not repay their redemption value in full). In other words, if the loan portfolio is subject to significant potential or incurred losses during the year, the Cost of Risk rises accordingly, therefore reducing the NBI.

The P&L (before tax) is broadly equal to the NII less the costs (e.g., employees, IT, buildings ...) and if positive, the P&L (after tax) rewards the shareholders. One of the main building blocks of the banking regulation is to impose to the shareholder(s) to mobilize a **minimum amount of capital** to sustain the risks borne by the bank and, all other things remaining equal, in some instances even raise additional amount of capital thereby lowering the Return (P&L) on Capital⁴ (Equity) for the shareholder. Recall that **in case the P&L of the year is negative (loss), it reduces the amount of Equity accordingly, and may place the bank in trouble in case the remaining capital becomes close to the minimum level of capital imposed by the regulators (see next sections).**

In the next section, we will describe the main risks underpinning the banking business and how they impact the P&L and affect the capital and the liquidity.

1.2 Overview of the main risks underpinning the banking business

This section aims at presenting in a simple and most intuitive manner some of the key risks run by banks and how these risks affect and even jeopardize the bank viability if they are not managed properly.

⁴ We do not distinguish Capital from Equity at this stage, as the former is the terminology specifically used by the Regulators as opposed to the latter which is more legal.

For the sake of clarity, we go back to the paradigmatic balance sheet of a retail bank, recall that this bank has a basic activity which consists in taking on client (sight, term) deposits and granting loans. This bank is thereby said to be “**refinanced**” (customary term) through client deposits.

1.2.1 Introduction to credit risk

The first intuitive and even obvious risk borne by this bank occurs when some clients start to not repay the interest and / or the nominal value of the loan and in that case, we say that the loan becomes “**non-performing**” or **enters into a state of default (“defaulting loan”)**. The risk of partial redemption (or no redemption) of the loan nominal is called is called the “**credit risk**” and, from the regulators’ viewpoint has always been the key risk which could trigger a bank failure (i.e., bankruptcy) in case a certain number of loans would default.

Indeed, in case the loan becomes non-performing, prevailing accounting rules impose to **estimate** the **loss** expected to be **actually incurred on the loan** and increase the **CoR** accordingly thereby reducing the NBI. In case the CoR exceeds the NBI, leading hence to a negative P&L, the Equity (shareholders wealth) is reduced. Of course, any bank will encounter non-performing loan, as the credit risk is consubstantial to the bank business and the bank makes profit by taking on **credit risk**.

However, up to which point a bank can withstand credit losses? If you place yourself in the shoes of a depositor (i.e., a client who deposited his cash in the bank – creating a liability from the bank’s viewpoint), up to which point will you accept the bank **to incur losses** on the credits granted with your deposits, as the primary bank business consists in transforming the client deposits into loans?

When a bank takes on credit risk by transforming client deposits (some of them may remunerated and some not, depending on the bank’s refinancing strategy) into loans, banks’ shareholders expect to **be remunerated against the credit risk** and to absorb the potential loans’ default and subsequent losses, thereby protecting all their clients’ deposits. Otherwise, the bank would never find any depositor willing to leave their savings knowing that they will be transformed into loans, a fraction of which may become dodgy.

Let us adopt the following notations to illustrate the situation:

- **N** denotes the nominal of a loan, with a maturity **T** and a bullet amortization scheme for the sake of simplicity. **r** denotes the fixed interest rate.
- $\widetilde{V}_T$ is the random variable denoting the value of the loan at maturity.
- V_T is the value of the portfolio at time T in **absence of default**, namely $V_T = N + \sum_{i=1}^T N * C$, where the redemption value is equal to the initial Nominal and all the interests received over the life of the loan.

We then have: $\underbrace{(\widetilde{V}_T - N)}_{\text{gain / loss}} = \underbrace{(V_T - N)}_{\text{gain in absence of default}} - \widetilde{L}$, where $\widetilde{L}$ denotes the random loss amount affecting a loan.

When the default occurs during the life of the loan, the **incurred loss** is denoted $L^* = N(1 - R)$ to which the accrued interest(s) are due but not paid are added (we did not represent them in the formula to avoid overburdening the notations), and **R** denotes the **recovery rate** achieved over the nominal (and interests).

We rewrite this equation as: $\widetilde{L} = V_T - \widetilde{V}_T$. Therefore, when the bank is pricing the credit risk, it will charge to its client the **expected loss (“EL”) attached to the loan**, with: $EL = E(\widetilde{L})$, or a component of the total interest rate denoted $r^* = \frac{EL}{N}$ (loan **credit spread** – see below). The expectation is computed under the historical probability, but we will purposely not elaborate any further with regards to the statistical models used in this computation.

Under the IFRS, at loan inception (i.e., $t=0$, when the loan has just been granted) the nominal of the loan is reduced by the EL and (“double entry” principle) the Equity is reduced by the same amount. This feature is prudent as it illustrates that “statistically on average” it is expected that the loan may not fully redeem.

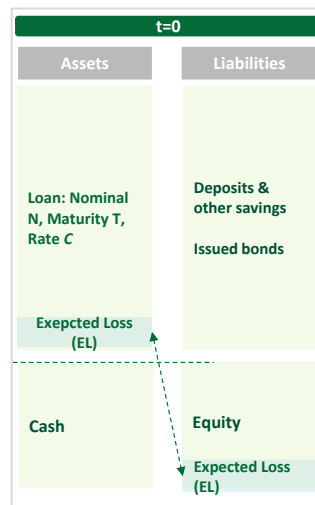


Figure 4 –Expected Loss at $t=0$

However, if the EL is deducted from the Equity, it may be erroneously interpreted without further precision that the Expected Loss is supported by the shareholders. In that case, the Equity would be reduced by the EL each time a loan is granted, soon becoming to a state where it would be close to zero and synonymous of a bankruptcy. Obviously, this reasoning is incomplete as the EL is supported by the client himself. Then, the total interest rate charged to the client should be equal to: $C = r^0 + r^* + r^c$, where:

- r^0 denotes the “risk-free rate” (or at least the cost at which the bank remunerates the depositors and/or the bond holders-if any), in other word, the minimal cost of the resources for the bank.
- r^* denotes the **EL (expected loss) component** (and should overtime offset the expected loss accounted at loan inception).
- r^c denotes the **commercial margin (or commercial spread)** which remunerates all the costs (e.g. employees, building, IT...) and ultimately the shareholders.

As a definition, r^* is also called the **credit spread**. In practice, we observe $r^* + r^c$, without any possibility to distinguish the credit spread from the commercial margin and by abuse of language the quantity $r^* + r^c$, may be denoted as credit spread.

In other words, the margin on this loan is equal to: $N * C - N * (r^0 + r^*) = N * r^c$, as $N * r^0$ remunerates the Ressources (client deposits, bondholders) and covers the losses if the **incurred losses (IL)** $L^* = N(1 - R)$ equal the expected losses computed at loan inception (see Outcome #1) $L^* = EL$, then the credit spread paid by the client and accounted as a profit at $t=T$ has fully compensated (offset) the reduction of the Equity accounted at loan inception.

It also is obvious, that if at $t=T$ if the incurred losses are less than the expected losses, the bank accounts for an additional profit (and a greater Equity amount).

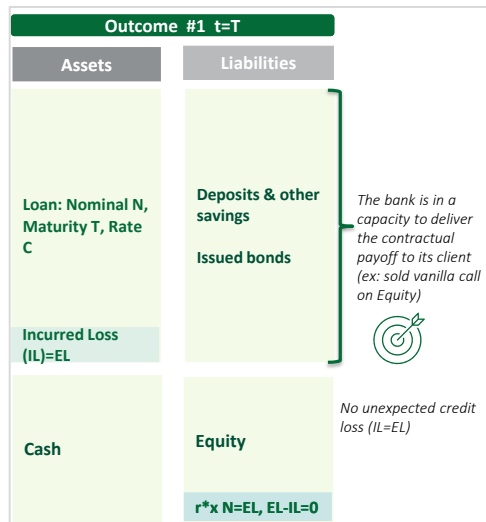


Figure 5 –Outcome #1: $EL=Incurred\ Loss\ at\ t=T$

The key argument underpinning the whole reasoning is related to the determination of EL or equivalently r^* . We asserted that it was determined through a model, subject to a certain number of assumptions which may turn out to be inappropriate if an **unexpected credit crunch** would arise (e.g., higher actual probability of default than that historically observed, lower recovery rates ...). Besides, prior to granting a loan, there is a decision (based mostly on scoring models) to grant or refuse a loan; again, if the granting process is flawed (e.g., erroneous scoring model) **then the incurred losses could exceed by far the expected losses and therefore trigger massive negative NBI and thereby a negative P&L entailing a reduction of the Equity.**

The positive difference between the IL and EL ($IL-EL>0$) is called the “**unexpected loss**” (UL).

If a depositor starts to believe that the P&L will be so negative (i.e. that the potential / incurred losses are so significant) that it will eat up, say, half of the equity, the same depositor(s) will start acknowledging that their deposits have been lent to non-creditworthy clients and that, in the coming years, the cumulated incurred losses will **exceed the equity** and start to crunch the amount of their own deposits (i.e. depositors will not recover the totality of their savings). When a depositor (or a sufficiently large group) of depositors, has any hint that such a situation may occur (through some news spread by the media and pointing at the difficulties of a given bank), those depositors will endeavor to withdraw their deposits (especially the sight deposits) triggering hence a “bank run” and the immediate bankruptcy of the bank because of its inability to repay the deposits. The risk run by a bank of not being in a capacity of **repaying its own debt** (when there are due and claimed, or simply not having the capacity to pay a cash outflow due and claimed) is an essential component of the **liquidity risk**. Recall that legally, the inability of any company (not only a bank) to not repay a debt past due and claimed entails an immediate bankruptcy.

If the Equity is *sufficiently significant* to absorb the UL (see Figure #6) without triggering a bank run from depositors willing to withdraw their savings, then the bank remains afloat even under extreme conditions, i.e. when unexpected losses occur. In other words, the shareholders should invest a *sufficient* amount of equity which they may completely lose in case the banks are faced with unexpected credit losses.

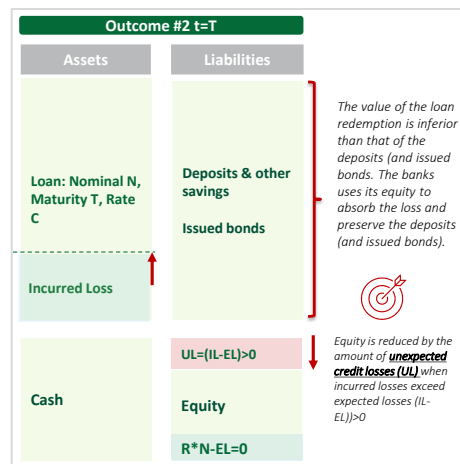


Figure 6 –Outcome #2 : $UL = IL - EL > 0$

How to prevent such a situation from occurring? First, by implementing strong processes within the bank to assess the credit worthiness of the client the bank contemplates to grant a loan to. Second, by ensuring that the equity amount set by the shareholders is in a capacity to **absorb the credit losses so that the depositors would “almost surely”⁵ never bear any loss (even under stressed situations)**. The banking regulation aims at setting this minimal level of equity (also called capital) to sustain unexpected losses.

Third, to prevent a bank from lending all their funds (e.g., client deposits) and impose them to conserve a certain amount of cash on their balance sheet to repay a certain amount of (sight) deposits and prevent a bank run. We will see thereafter the role of the shareholder’s equity to cover the credit risk.

1.2.2 Introduction to market risk

In case we have a bank engaged solely into CIB activities, more precisely derivatives and/or securities **trading activities**, it may also be subject to losses on derivatives and securities portfolios. You may have come across derivative valuation during your past tenure and you may remember that for optional & non-optional derivatives the **value** is determined through a **model (built on a set of assumptions) and calibrated on market parameters**; moreover, for optional derivatives, the value of the hedging portfolio is equal to the expected pay-off (under the risk neutral probability, assuming complete markets).

Nothing permits to assert that the model assumptions remain valid in any market configuration. In the same vein, the calibration methodology embeds also some uncertainties, and sometimes market quotes are not always available (e.g. implicit volatility for deeply in-the money options, continuous delta-hedging on options, interest-rate swaps for long term swaps in exotic currencies).

All these sources of uncertainties define the **market risk**, namely, the risk borne by the bank when running the trading activity and stemming from the fact that once the bank has **set a derivative’s price** (e.g. the option premium, the fixed rate leg of an interest rate swap) under certain market conditions prevailing at inception, it is committed to pay the contractual pay-off of the derivative. In case the hedging strategy becomes inefficient due, for instance to poorly designed models, unexpected very volatile market conditions affecting the underlying or even a lack of calibration prices (e.g., for deep in/out of the money options, long term interest rate swaps), **the value of the hedging portfolio may become inferior to that of the contractual pay-off** entailing hence a trading loss.

⁵ The careful reader would notice the probabilistic reference purposely laid down.

Let us adopt similar notations as those for the credit risk to pose the problem **for an option** (but a similar reasoning could be applied for a portfolio of non-optional derivatives such as interest rate swaps or even for a portfolio of securities composed of long and short positions):

- Let $\widetilde{C}_T$ be the random value of the option and $\widetilde{H}_T$ the hedging portfolio at time T (option maturity).
- We know from the financial theory (replication principle), that the theoretical value of an option at $t=0$ is equal to that of its hedging portfolio (under the risk-neutral probability), namely: $C_0 = \mathbb{E}^*\left(\frac{\widetilde{C}_T}{(1+r)^T}\right) = H_0$.
- However, as the value of the option is derived from a model, there are **uncertainties** which should be taken into account in addition to the theoretical values of the option, customarily denoted as **“reserves” (e.g., model reserves, calibration reserves)**. If we denote by $\widetilde{U}_T$ the random variable encompassing all those reserves, we then have: $C_0 + U_0 = \mathbb{E}^*\left(\frac{\widetilde{C}_T + \widetilde{U}_T}{(1+r)^T}\right) = H_0 + U_0$.

Therefore, the price paid by the client at inception ($t=0$) amounts to: $C_0^* = M_0 + U_0 + C_0$, where M_0 denotes the commercial margin (profit at inception) covering the fixed costs (trader salaries and bonuses, building, IT) and shareholder’s profit. In case the option is very complex, accounting rules also impose to recognize the profit at inception ($C_0^* - C_0 - U_0$) solely when the option matures; we will not delve any further in this direction.

t=0	
Assets	Liabilities
Positive Model Value (« Fair Value ») of the Hedging Portfolio: H_0	Negative market value of the (sold) option: C_0
Expected Losses (“reserves”): U_0	Expected Losses (“reserves”): U_0
Cash	Equity
	+ commercial margin M_0

Figure 7 –Market Risk at $t=0$

Likewise, we observe that U_0 is paid by the client also homogenous to an **expected loss “EL”** (under the risk neutral probability) on the hedging strategy; if we assume that we are on time $t=T$, we could come up with the following situations:

- $C_T = H_T$ and in that case, U_0 constitutes an additional profit for the bank (i.e., the initial uncertainties were too conservative).
- $C_T = H_T + U_0$ and in that case the additional price component embedding the uncertainties was necessary to undertake the hedging strategy without entailing a loss for the bank (see Outcome #1 on Figure 8), i.e., the reserves (EL) paid by the client were necessary to undertake the hedging strategy.
- $C_T > H_T + U_0$, and if we assume a sold option, the hedging strategy is not sufficient to compensate the payoff due to the client, entailing hence an **unexpected trading loss for the bank’s shareholders** (Outcome #2 on Figure 8).

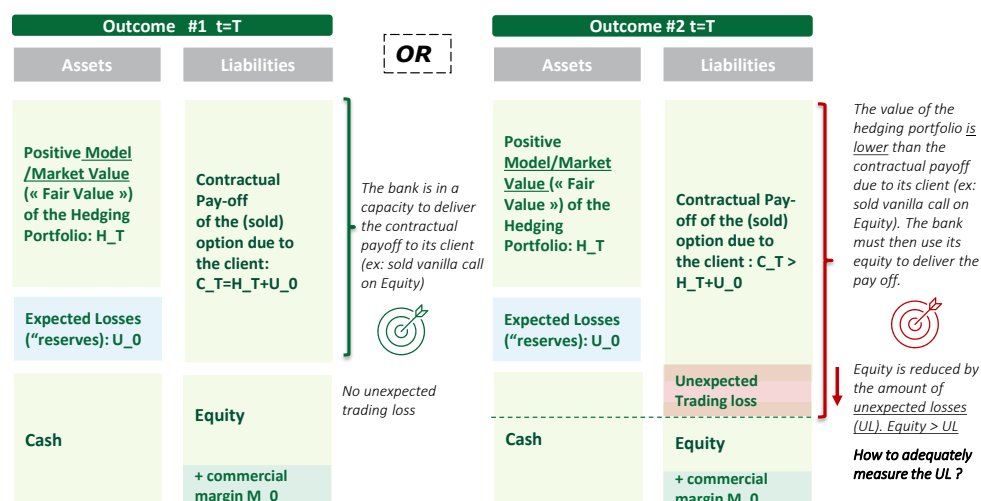


Figure 8 – Market Risk at $t=T$ for Outcome #1 and #2

In a similar manner as the reasoning laid down for the credit risk, we would say that the equity (i.e., shareholder's money) should absorb **unexpected trading losses** and that the debtors (bank's bondholders, if the bank issues securities to refinance itself) would almost surely never bear any loss. In that case, the losses reduce the shareholder's worth.

Therefore, we repeat and adapt the arguments laid down for the Credit Risk. Market Risk can be mitigated as follows: firstly, by implementing strong processes within the bank to carefully design pricing & hedging models and measure the uncertainties (EL) surrounding these models. Second, by ensuring that the equity amount set by the shareholders is in a capacity to **absorb the market unexpected losses ensuring that the depositors (universal bank) or bondholders would "almost surely" ⁶ never bear any loss (even under stressed situations)**. The banking regulation aims at setting this minimal level of equity (also called capital) to sustain unexpected losses for market risk. This minimal level comes as an addition to the minimal level dedicated to credit risk for those banks running credit and trading business.

As for a trading security (e.g., Government bond, equity stocks) activities, the same principles apply in a more simple fashion: the bank is committed to ongoingly buy and sell securities. The bank therefore holds an inventory of long and short securities on its balance sheet, and this inventory is evaluated daily, upon the market prices, and gain / losses are accounted in the P&L.

1.2.3 Introduction to (structural) interest rate risk

Let us go back to a simple retail banking business to illustrate how banks **structurally** bear an interest rate risk. Barring exception, whenever a credit institution operates its business by granting loans and taking on deposits or issuing bonds it bears an interest rate risk, hence justifying the adjective "structural" to illustrate that this interest rate risk lies at the heart of the banking business.

We consider through a "toy" example whereby a retail bank grants fixed rate loans refinanced by variable rate bonds (we purposely do not consider client deposits).

⁶ The careful reader would notice the probabilistic reference purposely laid down.

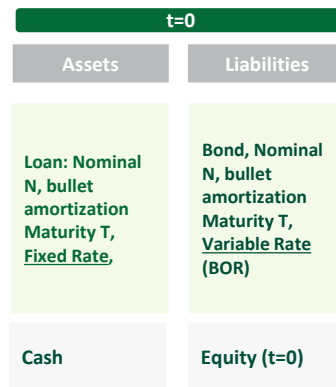


Figure 9 – Toy example: Fixed rate Loans refinanced by Variable rate bonds

For the sake of simplicity, this bank is assumed to solely have granted loans all bearing a **fixed interest rate (“FR”)**, with a bullet amortization profile (i.e. the nominal is redeemed all at once at maturity, assuming that all loans mature at the same date, namely 5 years). Recall, that a fixed rate is **contractually set until the loan matures**.

On the liability side, this bank has issued a variable interest rate bearing bond (of the same nominal as the loans) with a bullet amortization profile with 5 years of maturity; we customarily said that this bank is **refinanced** by a variable rate bond.

Recall that a variable rate bond is indexed on a variable interest rate reference (e.g. Ester, Euribor) which is said to ‘re-fix’ at a given frequency date. For instance, a 3-months tenor **Euribor (“BOR”)** fixed today and assumed to be paid in 3-month time will be set at a new level in 3-month time. If the interest rates rise, then in 3-month time the Euribor will be higher for the next 3 months than level fixed today; conversely if the interest rates fall, so does the BOR.

Let us now illustrate the stream of interest rate on the asset and liability side:

Years	1	2	3	4	5
Interest Rate Received (FR) from the Loan	FR	FR	FR	FR	FR
Variable Rate (BOR) paid on the bond	BOR	BOR	BOR	BOR	BOR
Net Interest Income (NII)	(FR-BOR)	(FR-BOR)	(FR-BOR)	(FR-BOR)	(FR-BOR)
Assumption : Euribor + 200 bps => NII :	FR-(BOR+200)	FR-(BOR+200)	FR-(BOR+200)	FR-(BOR+200)	FR-(BOR+200)
Assumption : Euribor - 200 bps => NII :	FR-(BOR-200)	FR-(BOR-200)	FR-(BOR-200)	FR-(BOR-200)	FR-(BOR-200)

Equity (t=0)

OR

Equity t=1

Equity t=2

Equity t=3

Equity t=4

Equity t=5

Equity t=1

Equity t=2

Equity t=3

Equity t=4

Equity t=5

Figure 10 – Impact of a stress on the NII

We observe that for each of the upcoming years, the NII is totally dependent on the level of the Euribor. If the Euribor rises by 200 basis points ⁷(“bps”), the NII is reduced by 200 bps and then all other things remaining equal so is the NBI and therefore the P&L, and if this latter becomes negative, it turns out to be a loss (reducing the Equity). Conversely if the Euribor falls by 200 bps, then the NII rises by 200 bps and,

⁷ 1 basis point (« bp ») = 0.01%

if other things remain equal, would increase the P&L and the Equity (and finally the shareholder's net worth).

Recall that losses reduce the Equity as opposed to profit which increase the Equity. It is also worth noting, that the structural interest rate risk is independent from the Credit Risk and Market Risk

In both cases, the upcoming P&L becomes unpredictable, not to say potentially volatile and completely dependent on the interest rate moves. Is this a desired situation for the bank CEO who is accountable to the shareholders? Let see the next paragraph!

In a more generic fashion, we say that when a fixed rate (resp. variable rate) element of the balance sheet at a given point in time is **refinanced** through a variable rate (resp. fixed rate) element of the balance sheet, that the bank is facing **an interest rate gap**. The risk management of the interest gaps is owned by the Asset & Liability Management ("ALM") function. Barring exception, ALM lies at the core of the banking business, as interest gaps are consubstantial to the banking business, thereby explaining once again the denomination "**structural interest rate risk**" outlined in the title of this section.

Therefore, the bank's CEO is not in a comfortable position if he purposedly leaves "open" the interest gaps as it is equivalent of running a speculative view on the interest rate level which could end up by increasing or reducing the value of the Equity. One way to mitigate this structural interest rate risk is to transform an interest rate gap (on a given nominal) into the opposite position; do you know a financial instrument bearing such features⁸?

If we go back to the balance sheet of the retail bank we presented at the very beginning of this document, we observe that the loans are refinanced by client deposits. If we assume that the loans are all fixed rate (e.g., in France this is customary for residential mortgages), and that the deposits all sight deposits (withdrawable at any time) and not remunerated (i.e., bearing a 0% fixed interest rate, also customary in France), do we still have an interest rate gap? The naïve answer would be "No", as fixed rate loans are refinanced by fixed rate (0%) deposits.

However, sight deposits could be withdrawn at any time and loans have a contractual maturity. Then **assuming** that all deposits are called tomorrow (T+1) is likely to be too conservative and would require the bank to find additional refinancing sources (fixed or variable rate) with a longer maturity to "survive" over T+1. Conversely, assuming that all sight deposits will remain unchanged until all the loans mature, is also naïve. ALM, therefore necessitates to design behavioral models to represent the dynamics of sight deposits (also denoted **non-maturing deposits**).

The regulators started to pinpoint and streamline the structural interest rate risk in 2015 by imposing certain limits⁹ on this risk and on the models used by the ALM to ensure that interest rate gaps were properly measured and hedged with the appropriate instruments. Additional constraints have been proposed in 2022 and enter into force in 2023. This is beyond the scope of this document and we will not elaborate further; you will address this subject during your ALM lecture.

1.2.4 Introduction to liquidity risk

Let us stick to the two previous examples whereby all loans are bearing a fixed interest rate with a bullet amortization profile are financed by a bond bearing the equivalent characteristics (same nominal and maturity date, let alone the interest rate which should be higher for the loans so as the NII remains positive and entails a profit for the bank).

If the Nominal amounts to 100 and the maturity is 5 years, we observe that there is:

- i) **a cash outflow** of 100 when the bank repays the bond at T=5 years
- ii) **a cash inflow** of 100 when the loan redeems at T=5 years

⁸ Answer: the Interest Rate Swap is a perfect candidate

⁹ Interest Rate Risk on the Banking Book ("IRRBB")

In this case, the cash inflow matches the cash outflow at $T=5$ years and there are not intermediary cash inflows or outflows.

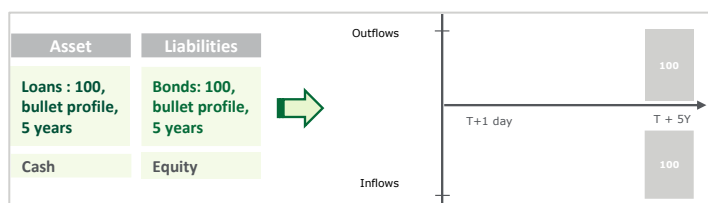


Figure 11 – Liquidity Risk – example #1 – cash-flows

Alternatively, if instead of representing the cash inflows and outflows, we present in an equivalent fashion the loans & bonds **outstanding nominal** amounts, we come up with the following illustration:

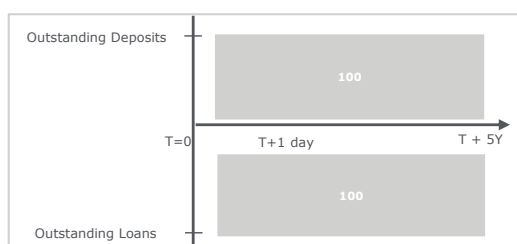


Figure 12 – Liquidity Risk – example #1 - outstanding amounts

In the second example, when the loans are refinanced by sight deposits (i.e. contractually callable at any time), without posing **further assumptions** and sticking to the **contractual clauses of the deposits**, we have:

- i) **a cash outflow** of 100 at $T=1$ day
- ii) **a cash inflow** of 100 when the loan redeems at $T=5$ years

In this case, the cash outflow at $T=1$ day does not match the cash inflow occurring at $T=5$ years, it is said that the **net outflow** at $T=1$ day amount to 100. Recall that if this situation actually occurs in practice, in other words the bank cannot repay the called deposits, it triggers an immediate bankruptcy (“cease of payment”, pursuant to the legal denomination).

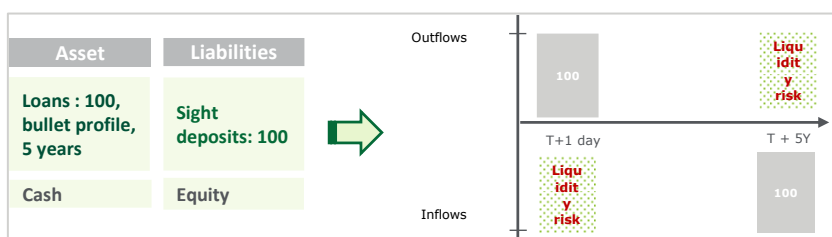


Figure 13 – Liquidity Risk – example #2 - cash-flows

Alternatively, if instead of representing the inflows and outflows, we present in an equivalent fashion the outstanding loans & sight deposits amounts, we come up with the following illustration:

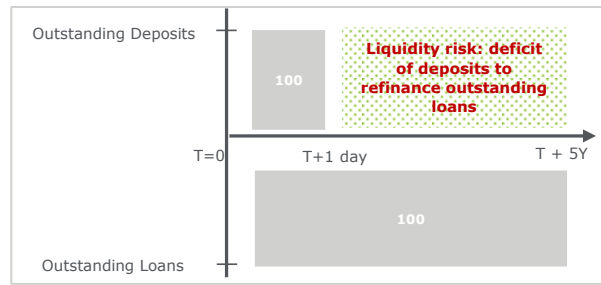


Figure 14 – Liquidity Risk – example #2 - outstanding amounts

If we now assume that the sight deposits amortize linearly over 5 years (for a deposit of 100 it entails an amortization of 20 per year), we observe that the difference between cash outflows and cash inflows still exist, but gradually spans over 5 years.

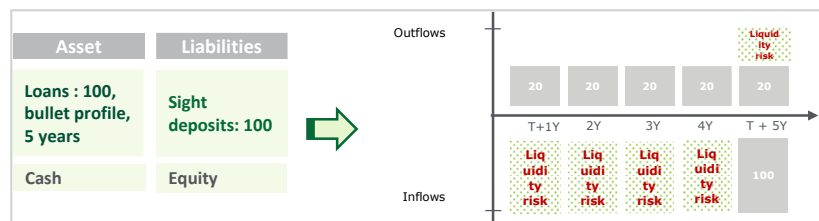


Figure 15 – Liquidity Risk – example #1 - cash-flows

Alternatively, if instead of representing the inflows and outflows, we present in an equivalent fashion the outstanding loans & sight deposits amounts, we come up with the following illustration:

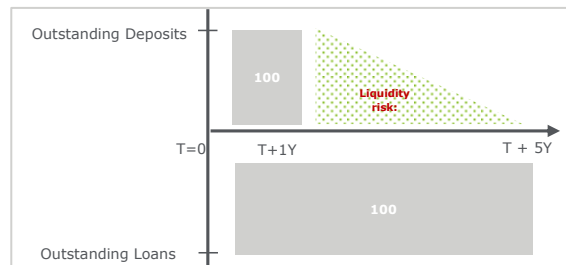


Figure 16 – Liquidity Risk – example #2 - outstanding amounts

As a summary, **we define the liquidity risk** as a situation where **the cash outflows do not match the cash inflows or equivalently if the net cash outflows or net cash inflows are not equal to zero**; we also speak of “**liquidity gap**” in the same vein as the liquidity gap previously addressed in this document.

Obviously, the critical situation occurs when the bank is faced with a net cash outflow which may trigger a bankruptcy situation in case the bank does not find appropriate refinancing. We will see later in this document, how to mitigate the liquidity risk and how the regulators imposed some constraints to mitigate it.

We also observe that, throughout this section, the loan is never subject to any default and we solely consider its full nominal as the surrender value (without considering the interest due). Recall also that accounting and economic wise, the redemption of a loan has no impact on the P&L (as the bank’s client solely repays the nominal amount lent); solely the interests received from the loan and those paid to the depositors or the bank’s bondholders have a P&L impact. In case the loan becomes non-performing, its nominal is reduced by the amount the bank expects to not recover (or expects to lose).

1.3 Summary: role of the Capital

In the previous section, we have explained in a sketchy albeit intuitive manner that, as long as the equity amount (also called capital) is sufficient to cover the **unexpected losses** attributable to credit risk and/or market risk, then the depositors would *almost surely* not lose their savings. Recall, that the **expected losses** are passed on to the bank's client through the pricing mechanisms (**credit spread** added to the risk-free interest rate for a loan, various type of reserves added to the derivatives' model price).

The figure #17 illustrates that the equity amount mobilized by the shareholders should be greater than the unexpected losses, as the shareholders primarily absorb the losses.

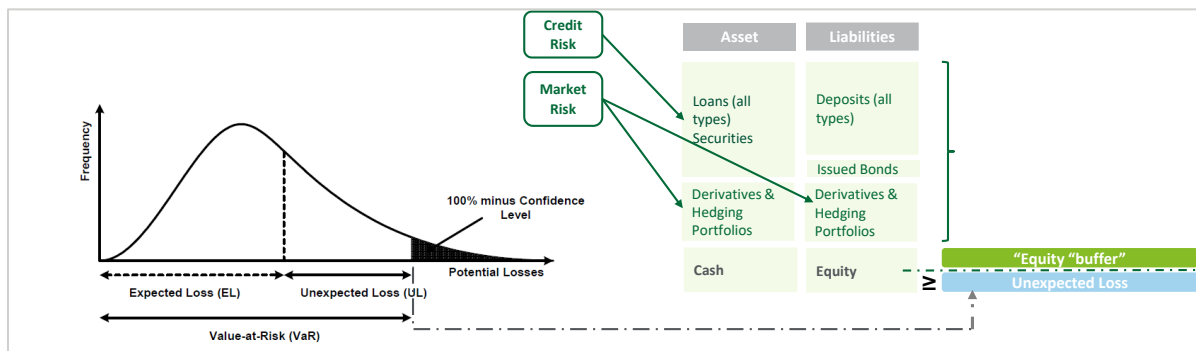


Figure 17 – Role of the Equity / Capital to absorb UL

Once said that, there are still key questions to be addressed:

1. When we speak of unexpected loss or alternatively, we say that the depositors would *almost surely* not lose their savings, what is a sufficient level of confidence to impose? In other words, shall we say that the depositors would not lose their savings in 97%, 99% or even 99.5% of the time?
2. In the same vein, to which time horizon shall banks measure their unexpected losses (e.g. 1 year or the maturity of the longest loan or derivative contract)?
3. Which technique(s) should banks use to measure their unexpected losses? How do the supervisors ensure that banks use appropriate techniques to measure their unexpected losses? We observe in the figure above that statistical techniques dedicated to extreme-value measurement should be used.
4. How the regulators ensure that those risk measurement techniques are sufficiently streamlined across banks and/or between banks operating in different countries and ensure comparability and efficient supervision. How to limit moral hazard from banks' senior management and ensure a level playing field (i.e., limit distortion in competition for those banks willing to use more "aggressive / less conservative" risk measures)?

We address these questions in the next section while introducing the Basel agreements and their translation into the European regulatory regime when it comes to equity / capital measurement. The specific risk measurement techniques will be elaborated more in-depth in the second part of this lecture. The interested reader may consult [1] – Chapter 29 – section 5 for a discussion on the **Regulatory and Economic Capital**.

1.4 How to cover the liquidity risk(s)?

In contrast to Equity (capital) which intends to absorb losses attributable to credit and/or market risk, we did not see any elements available to cover the liquidity risk. Nevertheless, there exist some mechanisms and constraints which may be imposed onto the balance sheet structure to reduce liquidity risk.

Prior to going any further, we distinguish the “liquidity” from the “funding” risks which are both used in practice but not in the same spirit. **Liquidity risk** refers to short-term horizon, namely when cash outflows do not match inflow up to 1 year and where the banks need to find resources on the short run (e.g. borrow cash from another bank for – say – 2 days- on the interbank market). **Funding risk** refers to the same mechanism from 1 year to the full amortization of the bank balance sheet. Barring exception, we will stick to the term liquidity risk in the remainder of the document to designate both liquidity and funding risk.

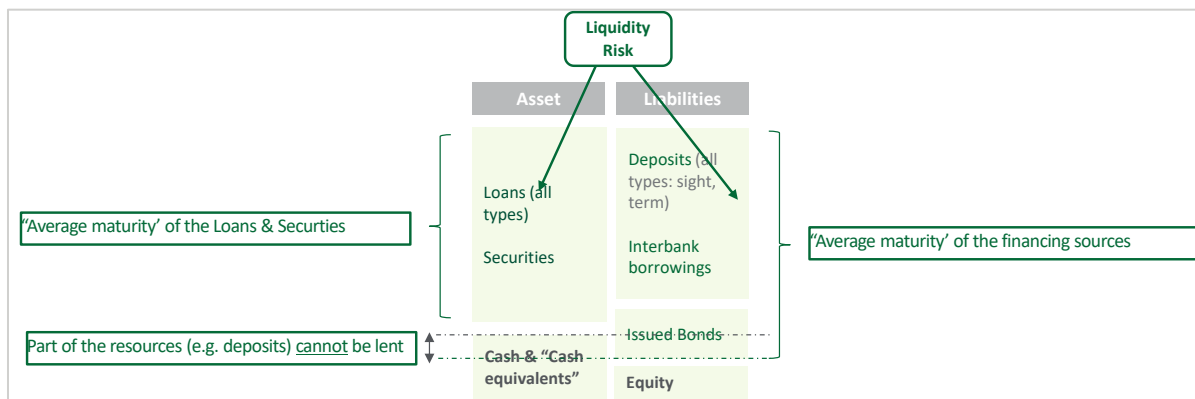


Figure 18 – Constraints to mitigate liquidity & funding risk

The first mechanism to mitigate liquidity risk and avoid the situation exposed on Figure 13 whereby a 5-year bullet loan is refinanced by sight deposits would consist in limiting the bank in its capacity to lend short term resources on the long run and by extension to ensure that the *average* maturity of its resources (e.g. sight deposits, issued bonds) is equal or larger than the *average* maturity of its assets.

At the first glance this constraint appears obvious, but it should be calibrated in a certain way to not destroy the banking business. Namely, in an upward sloping interest rate curve environment ¹⁰, bank borrow their resources on the short term (as they are cheaper) and lend them on the long term to benefit from the **interest rate transformation** which lies at the heart of the banking business.

Imposing a constraint in which a bank should have the *average* maturity of their liabilities equal or exceeding that of its *assets* may place the bank in an unsustainable “**anti-transformation**” position, whereby the cost of its resources would exceed the return on its assets. The constraint imposed on the determination of the **average maturity** should be carefully calibrated to avoid the “anti-transformation” position whilst mitigating the funding risk (we purposely used the funding risk terminology as we are dealing with a long-term horizon). In particular, the amortizing scheme on client sight deposits should not be too severe and more in the vein of that exposed on Figure 15 and 16. We briefly address this constraint denoted as the **Net Stable Funding Ratio (“NSFR”)** in the second part of this document.

The second mechanism to limit the liquidity risk consists in imposing a limitation on the resources that the bank may lend and impose that a certain percentage of the resources at risk (e.g., sight deposits which may be withdrawn overnight without notice period) should be set aside (e.g., deposited in the Central Bank) and never lent. Again, such a mechanism entails a loss of business as it curbs the bank’s lending capacity but enables the bank to heighten the cash at its disposal to face deposits outflows under a

¹⁰ This upward sloping assumptions holds to be true but not always and depends on the currency. Between 2015 and 2021 the Euro interest rate curve was almost flat.

situation of stress over a limited time horizon. The regulators imposed a constraint denoted **Liquidity Coverage Ratio (“LCR”)** that we briefly address in this document.

Against all the odds, we will see that the regulators did concentrate their efforts to streamline the adequate size and quality of the capital rather than focusing on liquidity which is in essence and in practice what triggers banks’ default. In other words, we say that “bank die rich” (i.e., bank die being “solvent” but not “liquid”).

2. Overview of the key features related to CRR (EU Basel agreement translation).

All the concepts exposed in the previous chapter and addressing equity (capital) requirements have been set into a series of principles and rules, those rules being enriched and complexified and named as **Basel Agreements (or Basel Accords)**. In this lecture, we will not delve into the history and complexities of the Basel Agreements but solely expose the most important features from a risk management perspective.

Basel Agreements are named in accordance with the decisions issued from the Committee dedicated to banking supervision and hosted in the premises of the Bank for International Settlements (BIS) located in Basel. The BIS hosts nine international organizations engaged in standard setting and the pursuit of financial stability through the Basel process. In essence, a country may designate a representative in the BIS¹¹ to discuss the Basel principles which aim to globally streamline the banking regulation.

It is worth noting that, **the Basel Agreements are not binding until they become translated into applicable laws and regulations** prevailing for a country and / or a specific zone. In the European Union, the Basel Agreements – today the latest “Basel IV” package is under finalization - have been translated into a series of **Capital Requirement Regulations and Directives (“CRR” and “CRD”** – CRR II and CRD V have entered in force and CRR III and CRD VI are under finalization – those being proposed by the European Commission and voted by the European Parliament – see [3] and [4]).

Worth noting that new Basel Accords / new voted CRR/CRD do not repeal the existing ones (CRR I and CRD IV) but rather add complements and corrections in the existing texts which does not facilitate their reading. We will speak of CRR / CRD in the remainder of the document and even as their denomination is a misnomer as they address not only the level of capital to cover the unexpected losses but also the liquidity and other aspects we will not address in this lecture. Recall that in this chapter we will use the word **Regulatory Capital** or simply **Capital** (without further distinction) instead of Equity. Even if these two notions are similar, Capital / Regulatory Capital is broader in certain instances than sheer shareholders equity (we will briefly make this distinction). We will also address **Economic Capital** later.

We will now present the key principles underpinning the Basel agreements / CRR-CRD, namely the “**3 Pillars**” and introduce some of their key features. The remainder of this chapter is organized in accordance with the 3 Pillars (even if the Pillar 3 will be addressed in an allusive manner).

¹¹ The interested reader may consult the BIS website which is a genuine goldmine ([Bank for International Settlements \(bis.org\)](http://bis.org))

3.1 The 3 Pillars

The figure #18 illustrates the 3 Pillar underpinning the Basel Accords and delineated in all laws & regulations (e.g. CRR) for the countries who have adopted the Basel Accords.

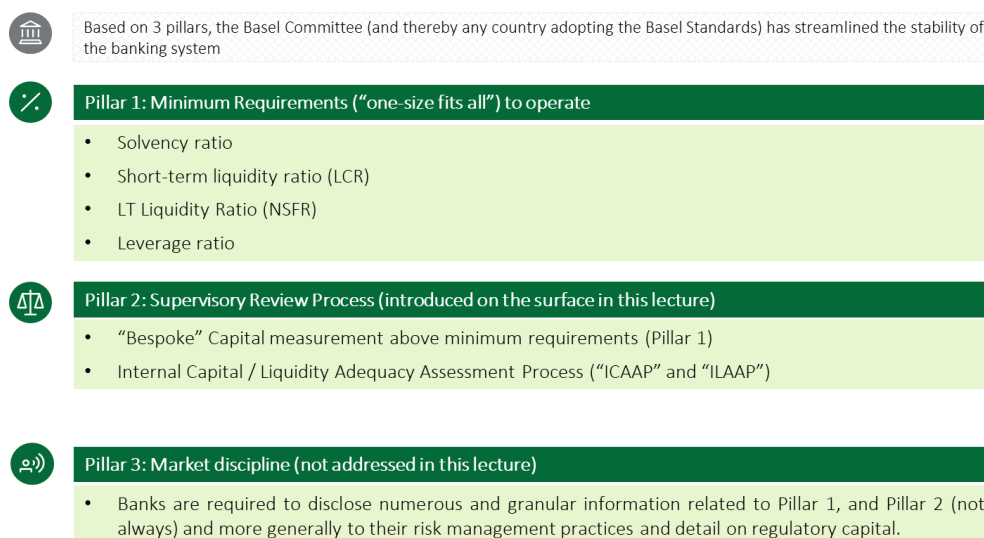


Figure 18 – The 3 pillars

The Pillar 1 sets the **Minimum capital and Liquidity requirements** applicable to all banks regardless of their size, activities, and complexity. In other words, **no bank** (or to use the adapted regulatory terminology, **credit institution** – or simply **institution**) is entitled to operate (e.g. take on deposits, grant loans, engage into trading activities) without complying with those minimum requirements. Therefore, any bank willing to operate in the EU is required to fill an application form which will be eventually validated by the European Central Bank (“ECB”) following a lengthy process spanning over 6 months calendar time as a bare minimum.

If we adopt a metaphor taken from the sartorial world, we say that the Pillar 1 is a “single size and ready-to-wear” suit applicable to all banks; obviously, such a suit cannot properly fit all banks (e.g. some of the risks may insufficiently be measured, some risks may be completely disregarded). Therefore, the Pillar 2 grants the power to the supervisory bodies to make some “bespoke adjustments to each suit”; in other words, the Pillar 2 tailors the Pillar 1 minimum requirements to a “new minimum” specific for each bank to take into account their own specificities.

Those “bespoke alterations” (denoted as “**Pillar 2 add-ons**” or simply “**add-ons**”) are determined in accordance with two processes respectively denoted **Internal Capital Adequacy Assessment Process (“ICAAP”)** and **Internal Liquidity Adequacy Assessment Process (“ILAAP”)**.

Be mindful that once the supervisors have set their expectations for the Pillar 2, those “new minimum”: i) cannot be lower than that of the Pillar 1 (i.e., it is never possible to taper the suit but solely to enlarge it), ii) become binding, i.e. the bank cannot operate below the Pillar 2. It entails that, in a competitive market, banks are willing to minimize their Pillar 2 add-ons as it requires to mobilize additional capital (Equity) all other things remaining equal (i.e., with a constant P&L/NII), leading hence to reduce profitability (also called “return on equity”).

The Pillar 3 is a series of **public disclosures** aiming at informing all stakeholders (e.g., investors, financial analysts, depositors) about the compliance with Pillar 1 (and lately with Pillar 2 – only for systemic banks in the EU) requirements and more broadly about risk management and internal controls. We encourage the interested readers to seek on the web the Pillar 3 disclosures (usually made of a comprehensive document ranging from 30 up to 300 pages) by typing in their favorite web browser “[Name of the Bank] Pillar 3”.

3.2 Main features from Pillar 1 - Capital

Let us come back to the series of questions we posed in section 1.3 to illustrate the design of the Pillar 1 when it comes to (minimum) capital requirements.

The regulators aim at setting standard level-playing field for the institutions operating under their supervision and ensure comparability across banks when it comes to measuring the minimum capital and liquidity requirements. If a regulator would allow each institution to adopt their own measures (e.g., compute their own loss distribution pursuant to Figure #17 and draw their UL estimates), supervision would be practically impossible as each supervisor would need to fully understand the methodologies, data and computation used by each institution to determine their UL. Besides, some small institutions may not have sufficient resources to hire a full-fledged team of risk managers and quants to perform such measures on an ongoing basis.

The Pillar 1 is then composed of a series of streamlined ratios bearing both on Capital (“Capital Ratios”) and Liquidity (“Liquidity Ratios”) to strike a balance between comparability across banks (to some extent), cost of implementation (especially for smaller banks) without compromising on banking stability.

3.2.1 Introduction to the Solvency Ratio

Let us start with the Pillar 1 **solvency ratio** (sometime called **capital ratio** – but today more of a misnomer as there exist another capital ratio called the **leverage ratio** as we will see thereunder) that should be complied with to operate as presented in Figure #19.

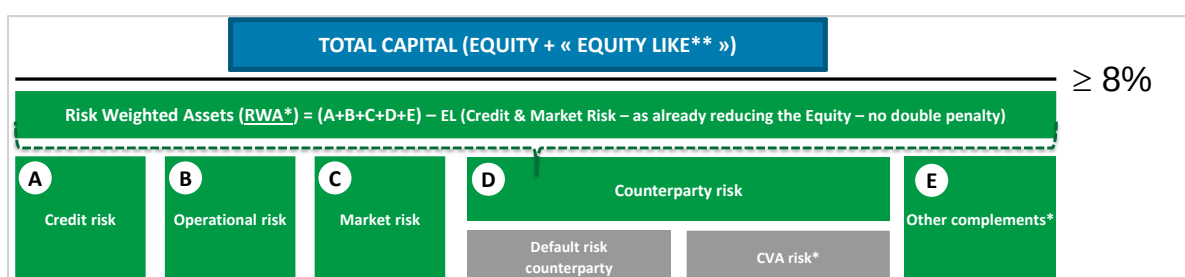


Figure 19 – Pillar 1 Solvency Ratio

The numerator amounts to the total available regulatory capital on the banks’ balance sheet pursuant to the Figure #17, whereby the denomination “Equity” was used as opposed to “Regulatory Capital” which encompasses Equity (i.e. shareholders money who own the bank) and also “Equity Like” (in Figure #19), i.e. **debt instruments** which from an economic viewpoint *behave* to a certain extent like equity (shares) but do not bestow their holders on any *ownership right* over the bank. The numerator is hence based on the accounting measure (book value of the shares – not the market value even in the case the institution is listed) and, if existing, the outstanding nominal of the “Equity Like” debt instruments.

We observe that the Pillar 1 capital ratio is **(arbitrarily) set by the regulators to 8%**¹². If we come back again to the denominator multiplied by 8%, this value is homogeneous to the Unexpected Loss on Figure #17. The difference lies in the fact that the UL measurement is not left to each institution, but streamlined by the ratio through the introduction of **Risk Weighted Assets (“RWA”)**; we will provide some details on the RWA but observe from the outset that the RWA is the *sum* of the following components:

- A. Credit Risk (already presented in section 1.2.1).
- B. Operational Risk (“Op Risk”) – we will briefly address this measure.
- C. Market Risk (already presented in section 1.2.2).
- D. Counterparty Risk – we will briefly address it.
- E. Other complements – we will purposely remain silent in this lecture.

¹² See [4] – Article 92

Regulators propose two approaches to measure RWA for each risk component:

- A **standardized approach** – mostly based on accounting figures multiplied by certain weights directly available in the CRR and therefore not necessitating the usage of any statistical techniques.
- An **internal model approach** based on statistical measurement techniques which requires a thorough validation (model, data, IT systems) from the supervisors to be authorized.

Institutions may use in conjunction standardized and internal model approaches across risk classes (e.g., internal model for Credit Risk and standardized approach for Op Risk). Barring exception, the usage of standardized approach and internal approach for a given RWA (e.g., Op Risk) is not permitted.

We will briefly see that through the definition of the RWA when using the internal model approach, regulators measure the **UL with 1-year horizon** and with a level of confidence of **99.9%** for Credit Risk, Counterparty Risk and Operational Risk and **97.5%** for Market Risk¹³.

Not to mention that the standardized approach is designed to be extremely conservative and comes up with higher RWA than the internal model approach.

3.2.2 Focus on certain RWA components – Credit Risk

Recall that **Credit Risk** is the risk of **loss** arising from the inability of the borrower to repay part / the totality of the **loan nominal** and / or past due interests. Recall again that the **Expected Loss** is invoiced to the client by adding a **credit spread** on the risk-free interest rate and that the **Unexpected Loss** (“UL”) is borne by the bank’s regulatory capital.

The measurement of the EL or UL necessitates to determine 3 components:

- The **Probability of Default** (“PD”) from the borrower, i.e., the probability that the borrower will not be in a capacity to repay the **outstanding (loan) nominal**¹⁴ within 1 year (this horizon is imposed by the regulators).
- The **Exposure at Default** (“EAD”) corresponding to the outstanding nominal (including the due interest rate) less (if applicable) any **protection** (e.g., mortgage, collateral, guarantee) in favor of the bank which reduces the nominal at risk.
- The **Recovery Rate** (“RR”) or in a dual manner the **Loss Given Default** (“LGD” equal to 1-RR) which simply measure the capacity of the bank to recover part (and sometimes the totality) of the EAD once the default has occurred.

As explained previously, the RWA on Credit Risk is measured through 2 approaches: i) the standardized approach (“SA”) or ii) the Internal Rating Based (“IRB”) approach itself split into two subcategories, “IRBF” and “IRBA” where “F” and “A” respectively stand for “Foundation” and “Advanced”. Once the RWA is computed, the equivalent UL attributable to Credit Risk is equal to 8% x RWA pursuant to the solvency Ratio and the regulatory Capital should at least be equal to 8% x RWA to make the bank solvent (i.e., authorized to operate).

Let us now briefly present each of these approaches and start with the SA approach which is as follows:

- EAD: the outstanding nominal is an accounting-based measure, and the protection value is also accounting based, albeit some (regulatory) standard coefficients mitigate their values (see next example).

¹³ Those statistical techniques used in the internal approach to measure Credit, Market and Op Risk will be presented more in depth the second part of this lecture.

¹⁴ Outstanding nominal equals to the initial amount borrowed less (if applicable) the partial repayment(s) of the nominal which have already occurred.

- The SA does not necessitate the determination of any PD and LGD and solely imposes the usage of some regulatory **Risk Weights**¹⁵ (“RW”) depending on the nature (e.g., Retail, Corporate, Sovereign State) of the borrower (also called counterparty).
- Then, we come up with **RWA (Credit Risk) = EAD x RW**.

Let us illustrate how the SA is applied through a simple, albeit realistic example. We consider a **residential mortgage loan** granted to a **retail client** whose outstanding nominal amounts to 100. The value of the residential property (protection) amounts to 60 thereby leading to an EAD = 100-60=40 to the retail client (i.e. 40 are not protected) and an EAD of 60¹⁶ on the residential property.

Then, the CRR imposes to use a RW = 75% for retail clients¹⁷ and RW=35% for residential mortgage, leading hence to a total RWA = 75%x40 + 35%x60=51 and a minimum of amount of capital = 8%*51=4.08.

Let us briefly move to the IRB approach which is (mostly) similar to the SA when it comes to EAD measurement; however, the IRB requires the institutions to determine their own PD for each nature of exposure (e.g. Retail, Corporates) and their **internal rating** and either impose a standard LGD (e.g. 45% or 60%) under the IRBF or permit the institutions to determine their own LGD under the IRBA.

Any institution willing to apply the IRB shall design its own **internal rating framework** which consists of a set of methods, processes, controls, data collection systems and information systems to assign each type of exposure (e.g. retail, Corporate) to a given tier or category and determine the attached PD and LGD (under IRBA) accordingly. Recall that the use of this approach requires prior validation from the supervisory bodies and prerequisites in terms of rating systems and risk quantification.

Once the PD are determined for each type of exposure, the CRR mandatorily requires the usage of the **Credit-Value at Risk**¹⁸ (or “Credit VaR”) to determine the RW, namely:

- $RWA = EAD \cdot RW$
- $RW = f(PD, LGD)$

$$RW = \left(LGD \cdot N \left(\frac{1}{\sqrt{1-R}} \cdot G(PD) + \sqrt{\frac{R}{1-R}} G(0.999) \right) - LGD \cdot PD \right) \cdot 12,5 \cdot 1,06$$

Figure 20 – Credit VaR formula

We observe in figure 20 that "G" is the inverse distribution function of the Normal distribution, and "R" is the regulatory correlation coefficient¹⁹ – not addressed here) and that the level of confidence is 99.9% (0.999) and the time horizon 1 year (you may not see it – unless you look at the Credit VaR mathematical proof). Observe also that 12.5 is the inverse of 8% so that in this very case the introduction of the RW is purely an artefact to encapsulate both the SA and the IRB under the solvency ratio; however, in that instance the Credit VaR directly measures the UL (the coefficient 1.06 has been introduced by the regulators for the sake of conservativeness without further ado).

Obviously, we cannot replicate the previous example of the SA onto the IRB as it would require the determination of the PD (1 year) and LGD for a retail client and that is left to the hand of each institution.

We will not delve into PD and LGD measurement which are grounded upon statistical/ econometrical techniques, but rather briefly examine how the RW are determined under the IRB approach. The

¹⁵ Regulatory Risk Weights means that they are provided directly by the CRR – see articles 107 to 134 addressing Credit Risk under the SA.

¹⁶ The CRR imposes to consider 60% of the residential mortgage value for the sake of conservativeness, but we purposely omit this requirement to not overburden the computation.

¹⁷ See [3] article 123.

¹⁸ Consult [3] article 154 to observe that mathematical formulas are present in regulatory texts voted by the European Parliament.

¹⁹ Which has no relation whatsoever with the “R” of the recovery rate – hence explaining the use of the LGD

subsequent part of this lecture will come back to the PD and LGD measurement as well as providing a mathematical proof for the Credit VaR formula which is derived under a certain number of key assumptions. The interested reader will also find a detailed proof and a comprehensive discussion in [1] – Chapter 29 section 3 and 4.

3.2.3 Focus on certain RWA components – Operational Risk

The **Operational Risk (“Op Risk”)** has not been considered in the beginning of this document but constitutes a typology which is included in the solvency ratio. The CRR defines the Op Risk as the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events, and includes legal risk.

Under CRR3, the computation of Risk-Weighted Assets (RWA) for Operational Risk is fundamentally reshaped, as the previous CRR2 approaches (such as the former Basic Indicator Approach-“BIA” and the Standardized Approach – “SA”) are fully replaced by a single, standardized and more risk-sensitive (i.e. based on risk measures and not accounting measures) framework.

Instead of relying on a simple revenue-based indicator, CRR3 requires banks to calculate operational-risk capital using the **Business Indicator-“BI”**—a harmonized measure of business activity combining interest, services and financial components—and the Internal Loss Multiplier (ILM), which scales capital requirements based on a bank’s **historical loss experience**.

The formula becomes: **Capital Requirement = BI × BI Component × ILM**, and RWA are obtained by multiplying this capital requirement by 12.5. Again, under the BI the UL is directly computed and not the RWA in a similar fashion as the IRB. This results in an approach that is more comparable across institutions, more reflective of each bank’s risk profile, and more closely aligned with supervisory expectations for operational-risk management.

Let us illustrate the BI through a simple numerical example, where the assumptions are as follows:

- 3-year average BI = **€1,200m**
- BI bucket coefficient = **15%**
- BI × BI Component = $1,200 \times 15\% = \text{€}180\text{m}$
- Average historical losses (Loss Component) = **€60m**

Then the $ILM = \ln(1 + 60 / 180) = \ln(1.3333) \approx 0.2877 \rightarrow$ transformed to calibrated $ILM \approx \textbf{1.08}$. Henceforth, the **OpRisk capital requirements** = $1,200 \times 15\% \times 1.08 = \text{€}194.4\text{m}$ and **RWA** = $194.4 \times 12.5 = \text{€}2.43\text{b}$

3.2.4 Focus on certain RWA components – Market Risk

As market risk measurement techniques will be exposed in great length in the next part of this lecture, let us provide an overview and articulation of the existing landscape as well as an insight in the upcoming landscape.

Recall that **market risk** is the risk of loss arising from adverse movements in market price affecting the trading business.

Current in force market risk methods aim at directly measuring the capital requirements (i.e., the UL) and not the RWA and are currently decomposed into two approaches:

- The **standard approach** based upon a multiplication of accounting-based measures (or simple sensitivities – such as duration) by a standard coefficient set (arbitrarily) by the regulator (see appendix for additional details).
- The **Internal Model Approach (“IMA”)** where institutions need to use a **market Value-at-Risk** approach (“Market VaR”) where, as opposed to the Credit VaR, there is no prescribed VaR formula relying upon a certain set of assumptions.

More precisely institutions opting for the IMA need to decompose each of their market risk sources (e.g. interest rate, FX ...) and compute their **capital requirements** by extracting a 97.5% quantile (i.e. 97.5% - 1 year market VaR). **Various market VaR approaches** may be contemplated (e.g., historical, Monte-Carlo) each of them bearing their own advantages and drawbacks. Nevertheless, as for the IRB, the usage of an IMA is subject to the authorization of the supervisor and institutions need to complete a comprehensive documentation to seek for approval. The next part of this lecture will come back to the various VaR approaches.

Under the IMA, we come up with: **Capital requirement for market risk = VaR + Stressed VaR**, where the Stressed-VaR is a 97.5% quantile extracted from a loss distribution built upon more severe losses than that of the VaR.

In addition, specific additional requirements apply: i) on securities portfolios subject to credit rating migration risk (**Incremental Risk Charge - IRC**), ii) on correlation portfolios (CDO Index & Bespoke) - Comprehensive Risk Measure. These two last requirements are based on a VaR measure measured on specific parameters (e.g. correlation parameter on CDOs).

In the coming years (2027), both the standard approach and the IMA will be repelled and superseded by the Fundamental Reform of the Trading Book ("FRTB") itself delineated into two approaches:

- The **Standard – Advanced Approach** which will be in essence similar to the Standard Approach and grounded upon sensitivities **multiplied by regulatory standard coefficients and added together derive the capital requirements** (see appendix for additional details).
- The **Alternative Internal Model Approach ("AIMA")** which will replace the Market VaR by the **Expected Shortfall**²⁰ ("ES") with the same level of confidence; you will observe in the next part of this lecture that the ES is more conservative than the VaR and constitutes what is called a consistent risk measure.
- In addition, as part of the AIMA, an additional requirement for **sudden default risk** will be added on the Equity and Credit portfolios. This calculation will be based on a VaR approach and credit risk hedging instruments may be recognized to mitigate capital requirements.

We observe that the level of confidence is 97.5% as opposed to the 99.9% for Credit Risk and that the UL (minimum capital requirement) differs between Credit and Market Risk; this observation is valid, however, mostly due to computational issues (and precision) for Market Risk. The regulators agree to add Credit Risk UL and Market Risk UL even if measured under two different levels of confidence (but both with a 1-year time horizon).

Again, the RWA attributable to market risk may be derived by multiplying the Capital Requirements by 8%.

3.2.5 Focus on certain RWA components – Counterparty Credit Risk

Recall that when a derivative displays a positive market value, it is accounted with an entry on the asset side of the balance sheet. In case the **counterparty** to the derivative (i.e., the institution's client) would default²¹, the bank would, similarly as for a loan, **claim the market value of the derivative** and try to recover part / or the totality of it. Observe, that in contrast to the loan business we speak of counterparty rather than borrower or client even if these notions do not differ in practice.

As such, any institution engaged in derivative trading is subject to **Counterparty Credit Risk ("CCR")** defined as the loss incurred by an institution on **its derivative exposures (i.e., derivative with a positive market value)** when one (or more) of its counterparties would default. Counterparty Credit Risk ("CCR") is in essence similar to credit risk, the only difference being that the derivative exposure fluctuates over time in accordance with the market moves. Therefore, the difference between CCR and Credit Risk

²⁰ See *An Axiomatic Foundation for the Expected Shortfall*, Wang Zitikis, in *Management Science* (2020) where the ES is characterized as a "consistent risk measure" in the sense you will discover in the second part of this course.

²¹ For instance, in 2008 when Lehman Brothers defaulted, its counterparties in the derivatives business displaying a positive market value (asset) from their perspective towards Lehman were placed in a dodgy situation.

remains in the determination of the **EAD**, all other features remain equal (i.e., RW for each counterparty are determined either under the standardized approach or the IRB with the same values as those used for Credit Risk).

The regulators propose 4 approaches to determine the EAD (see details in appendix):

- 3 approaches requiring minimal computation but each of them subject to certain conditions of application.
- The **internal model methodology** (“IMM”), computationally intensive and requiring prior approval from the supervisors.

Finally, the RWA for CCR amounts to: $EAD \times RW$ (either Standardized Approach or IRB). Recall, that any protection received to reduce the EAD is taken into account as for the Credit Risk. In the derivative business, such protections are denoted “margin calls”²² and are made of cash and / or bonds (usually of very high quality, such as EU Government bonds). It is worth noting that since the entry into force of the **European Market Infrastructure Regulation** (“EMIR”) in 2015, mandatory (daily) margin calls are imposed to most of the counterparties trading derivatives, hence reducing the CCR which has become today a subject of lesser importance for the banks.

3.2.6 Focus on the regulatory capital

On the figure #19, we stated that the numerator of the solvency ratio was composed of Equity and “Equity like” the sum of which is called without distinction **Capital** or **Regulatory Capital**. Let us come back on the definition of the various type of capital available for the institutions and provide with some additional details about the 8% solvency ratio.

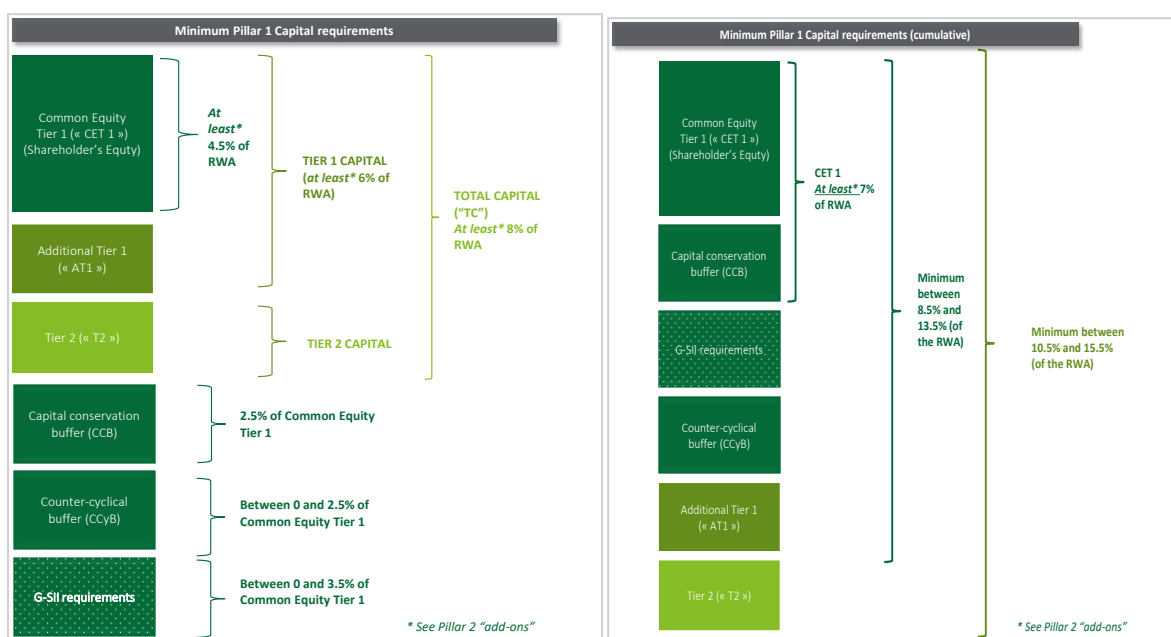


Figure 21 – Solvency Thresholds & capital typology

The Figure #21 presents the various type of capital available:

- Common Equity Tier 1 (“CET 1”) corresponds to the “genuine” Equity (i.e., Shareholder’s - bank’s owners money accumulated profit over the years) – this capital is the most solid one as it related to the owner’s money.

²² See [1] chapter 1 and 30 for a discussion on margin calls and more broadly financial markets organization.

- Additional Tier 1 (“AT1”) represents a **debt instrument** (does not bestow its holder on any ownership) whose features are similar to those of a share (e.g., no redemption date, subordination, discretionary coupon payment) – except the absence of ownership right over the bank.
- Tier 1 (“T1”) is simply equal to (CET 1 + AT1).
- Tier 2 (“T2”) also represents a **debt instrument** similar to the AT1 but more senior, i.e., in case of a bank liquidation following a bankruptcy the T2 holders are paid back before the AT1 holders. Not to mention, that the interest rate attached to the T2 is less than that of an AT1 (itself less than the return on equity expected by the shareholders).

We observe that the solvency ratio is split into 3 sub-ratios (4.5% for CET 1, 6% for T1 and 8% for total solvency ratio), in other words, it is prohibited to comply with the solvency ratio with 1% of CET 1 and 7% of Tier 2. Regulators impose to the bank’s shareholders to have “their skin in the game”.

Recall that these thresholds belong to the Pillar 1 and apply to all banks as a “ready-to-wear suit”. The Pillar 2 (through the ICAAP) may come up with **capital add-ons** specific to each bank and requiring higher binding solvency ratios. For example, if the supervisor following the ICAAP examination comes up with a capital add-on of 1% for the CET 1 for bank XYZ, the minimal binding capital requirement becomes 5.5% and not 4.5%. In case the solvency ratio for the bank XYZ falls below 5.5% the bank becomes insolvent and its license to operate will be withdrawn even if it still satisfies the Pillar 1 requirements.

On top of the solvency ratio, regulators also impose additional layers of CET 1 (without having a direct link with the UL), to foster banks to operate above their minimal Pillar requirements by building **capital buffers** as follows:

- **Capital Conservation Buffer (“CCB”)**: additional layer to the CET 1 to ensure that banks do not manage their business too close from the Minimum CET 1 – supervisors have the power to limit dividend distribution to restore the CCB.
- **Counter Cyclical Buffer (“CCyB”)**: additional layer to the CET1 built during “bullish” economic cycle and which could be used during “bearish” economic cycles.
- **G-SII requirements**: additional layer to the CET1 solely for the Globally Systemic Important Institutions (total balance sheet at least above EUR 30 billion).

However, in case of an economic turmoil, the additional CET 1 buffers set aside may be used by the bank to absorb **“losses exceeding the UL”** (thereby acknowledging the consubstantial imperfections of UL measurement models) provided that they remain above the minimal solvency ratio. For instance, in the EU, supervisors set the CCyB to 0% during the pandemic.

3.2.7 Leverage Ratio

Whilst revamping the Basel II accords (translated into the CRR as well) in the aftermath of the subprime crisis’ turmoil, supervisors were willing to impose an additional capital ratio which would be “model independent” (i.e., without standard or internal approach based on statistical measurement technique) to also foster comparability across banks.

Regulators hence came up with the Leverage Ratio (“LR”), purely accounting-based as presented in Figure #22.



Figure 22 – Leverage Ratio

In other words, T1 capital (and not solely CET 1) should be greater than 3% multiplied by the total assets and off-balance sheet items. Recall that off-balance sheet items may composed of various elements (e.g., guarantee offered by one bank to finance a loan at a later date upon request of a client, guarantee to ensure due payments on behalf of client if this latter would default). As for the derivative exposures, LR imposes a “non-IMM” method to permit comparability across banks.

The **solvency and leverage ratio** are commonly called **capital ratios** by the practitioners.

3.3 Main features from Pillar 1 – Liquidity / Funding

We have presented in section 1.4 the measures taken by the regulators to mitigate the liquidity / funding risk through the introduction of two ratios (NSFR and LCR) and we discussed their consequences on the banks’ profitability as they limit their transformation capacity.

We come back to these ratios and provide additional details related to their basic features. Even if the NSFR came into force (long) after the LCR, we start with the NSFR for the sake of clarify and for pedagogical reasons.

3.3.1 Constraints on Liquidity – Net Stable Funding Ratio

Regulators have imposed since 2019 to all banks a “ready-to-wear / one size fits all” ratio to cover **liquidity risk** on the long term (also called as **funding risk** when addressing horizons over 1 year). This ratio called **Net Stable Funding Ratio** (“NSFR”) aims to ensure that the assets are refinanced with liabilities displaying **on average** a greater maturity than those of the assets to avoid **on average** having cash outflows not covered by cash inflows.

In that context (see Figure #23 on the left-hand side), we say that the **assets require stable funding** (“RSF” – on the NSFR denominator) whilst the **liabilities offer an available stable funding** (“ASF” – on the NSFR numerator). Almost all assets and liabilities are eligible as RSF or ASF. The net stable funding ratio (ASF/RSF) should always be greater than 100%.

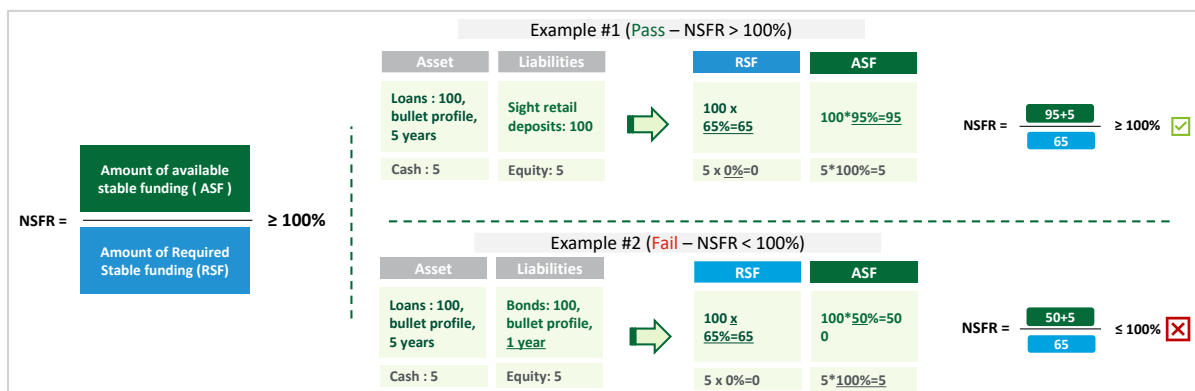


Figure 23 – NSFR and illustrative example

The Figure #23 illustrates across two examples the computation of the NSFR. The various regulatory weights ²³(e.g., 0%, 65%, 90%, 100%) are directly taken from the CRR (regulatory requirement) and depend on the nature of the assets & liabilities. Computing the NSFR solely requires mapping each accounting entry onto its corresponding regulatory weight. No internal approach exists for the NSFR. In the example #1, the institution is compliant with the NSFR, whilst in breach in the example #2.

We observe in example #1 that **sight retail deposits** (even if they may contractually be withdrawn overnight) attract a 95% ASF weight (and in some instances 90% – not detailed here), much favorable than a refinancing through a bond with a 1Y Maturity (50% weight) in example #2. The weights have been set by the regulators (and following a dialogue with the banks) in a manner to not over penalize the funding strategy through retail deposits which is in essence the core role of the banking business (i.e., collect deposits to transform them into loans, rather than borrowing fund under bond issuance on the capital markets).

It is also worth noting, that in the EU, the supervisors are currently not entitled to impose a **binding funding add-on for the NSFR** hence raising the minimum threshold above 100% under Pillar 2. However, in other jurisdictions (e.g., UK), the supervisor may require a binding add-on. In this context, any institution falling below the new minimal NSFR threshold would become “illiquid” and not be allowed to operate any longer.

3.3.2 Constraints on Liquidity – Liquidity Coverage Ratio

Regulators have imposed since 2014 to all banks a “ready-to-wear / one size fits all” ratio to cover **their liquidity risk on the short-term** and called the **Liquidity Coverage Ratio (“LCR”)**. In a similar manner as for the NSFR, the regulators require to identify all assets & liabilities (including commitments to receive or provide liquidity – and are accounted as off-balance sheet items) bearing a contractual maturity of less than 30 days; those assets & liabilities with a maturity above 30 days are not eligible as not posing any liquidity issue on the short-term.

The accounting values of the eligible items are multiplied by a regulatory weight provided in the CRR²⁴. The weights have been set by the regulators to account for a period of stress. Eligible assets are transformed into equivalent **cash inflows whilst eligible liabilities are transformed into cash outflows. The net outflows** (the LCR is designed in such a way – not discussed here – to always come up with a cash outflow by putting a floor on the inflow - purposely not presented here for the sake of clarity) represent the **denominator of the LCR** as seen on Figure #24.

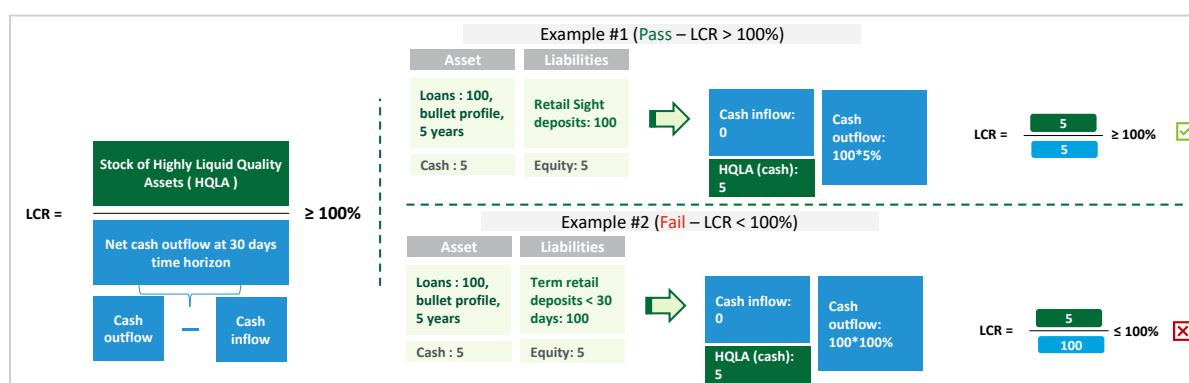


Figure 24 – LCR and illustrative example

A bank is required to satisfy a LCR above 100% and in that regard should hold a portfolio composed of **Highly Liquidity Quality Assets (“HQLA”, e.g. Cash deposited in the central bank, Government Bonds from EU countries)** to withstand the net cash outflows. In other words, the banks must hold a certain

²³ See [4] article 428

²⁴ See [3] articles 421, 422 and 425

amount of HQLA to face a potential bank run and avoid bankruptcy (“cease of payment”) by being able to repay the deposit withdrawals through the liquidation of the HQLA portfolio (even under stress as the assets composing the HQLA liquid and of very high quality).

We observe in the example #1 that the bank satisfies its LCR whilst being not compliant in example #2 as the regulatory treatment applicable to **sight retail client deposits** (5% of their amount) is more favorable than that applicable to term retail deposits with a residual maturity of less than 30 days. We also note that the 5% weight is consistent with the 95% weight applicable to the NSFR, namely if 5% of the **sight retail client deposits** are withdrawn within 30 days, it is equivalent to say that 95% of those deposits remain above 30 days and eligible as ASF. Besides, even if 5% outflows may appear not significant at the first glance, it applies over 30 days, and in a 1-year period of stress would cumulate to exceed 60%, occurring a massive decrease of the banks’ balance sheet.

Similarly, as for the NSFR, no internal approach exists for the LCR and, in the EU, the supervisors are currently not entitled to impose a **binding funding add-on for the LCR** hence raising the minimum threshold above 100% under Pillar 2. However, in other jurisdictions (e.g., in the UK), the supervisors may require a binding add-on which would raise the minimal threshold for the LCR and/or NSFR above 100%.

3.4 What is the Pillar 2

Recall that the Pillar 1 sets out the minimal capital and liquidity / funding requirements to respectively cover the unexpected losses attributable to credit/op/market/counterparty risk and mitigate the liquidity and funding risks. Those requirements are materialized through **two capital ratios** (the solvency and leverage ratio) and two liquidity ratios (the NSFR and LCR). However, those ratios and their applicable thresholds should fit all-banks regardless of their size, business and there is little chance that the Pillar 1 “ready-to wear” would capture all banking models.

In that regard, the Pillar 2 is a more bespoke approach whereby institutions are required to re-assess their risks **as if there were no Pillar 1** and compare the outcome of this assessment to the results of the Pillar 1. In that regard, institutions are free to choose the risks they are willing to include in the Pillar 2 as well as their respective measurement techniques.

If the capital needed to cover the **UL under the Pillar 2** exceeds that needed under Pillar 1, supervisors will impose additional capital requirements (called “**capital add-ons**” or simply “**add-ons**”) on the **solvency ratio** as already exposed in 3.2.6. The level of capital needed to cover the risks measure under the Pillar 2 is called the **economic capital** (or also internal capital) which is a misnomer it can never be strictly inferior to the Pillar 1 (i.e. capital add-ons cannot be strictly negative) even if the institutions demonstrates that it may be the case. The second part of this lecture will come back to the Economic Capital and present some measurement techniques used in practice.

In the EU, supervisors are not allowed to impose capital add-ons on the leverage ratio. As for the LCR and NSFR, institutions are required to perform a similar exercise to determine their own liquidity / funding risks under Pillar 2. In the EU supervisors are not entitled as well to impose liquidity add-ons for the LCR and NSFR but will put additional scrutiny to examine how institutions have assessed internally their liquidity / funding risks.

This section briefly illustrates how institutions in practice conduct their Pillar 2 assessments whose conclusions are shared at least on a yearly basis with the supervisor who is entitled to challenge the conclusions brought by the institutions. In case of disagreement supervisors may impose for instance his own view and add-ons on solvency and impose to seek for additional funding sources to strengthen the NSFR and any other relevant measure.

There is little detail on the Pillar 2 in the literature as this exercise is strategic and banks are not willing to share too much on such a subject. Worth noting that the regulators issued some comprehensive guidelines on Pillar 2 (see [7], [8] and [9]).

3.4.1 Pillar 2 on Capital – ICAAP

The **Internal Capital Adequacy Assessment Process (“ICAAP”)** constitutes the framework to determine the Pillar 2 for the capital and is articulated around 3 steps as illustrated on Figure 25.

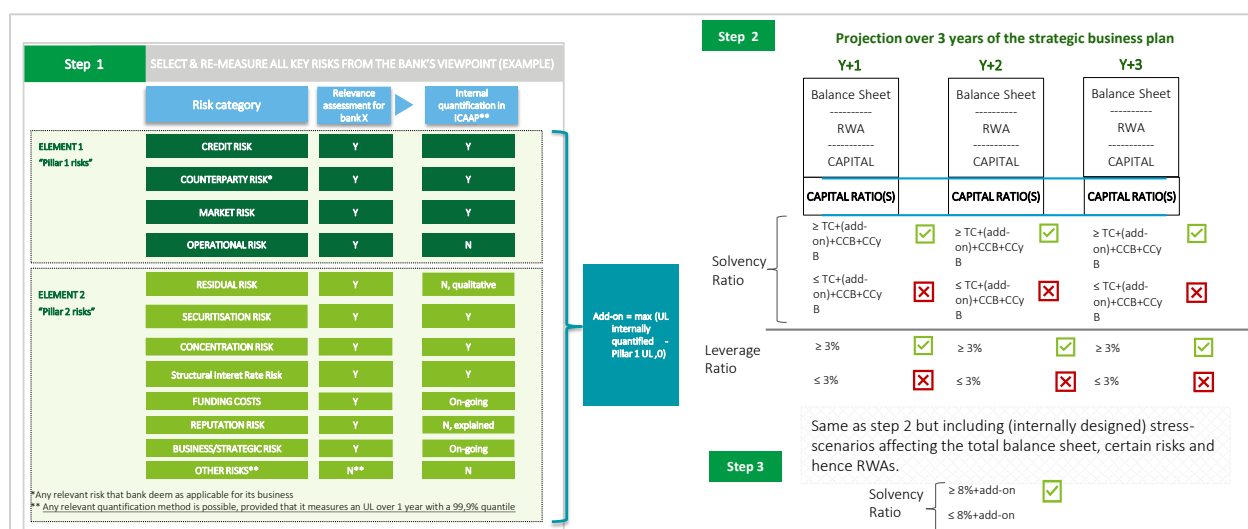


Figure 25 – ICAAP building blocks

The aim of step #1 is to ensure that institutions re-assess all their **material risks** beyond all those listed in the Pillar 1 (e.g., Credit, Operational, Market) – Figure 25 step #1 provides an illustration for the Pillar 2 risks. In that regard, it is worth noting that supervisors require that all institutions (independently of the ICAAP exercise) set up and maintain a comprehensive **cartography of all their risks** (financial and non-financial). To facilitate the work, there exist so-called **risk taxonomies**²⁵ which are streamlined frameworks on which institutions may map their risk to various granularity levels.

Once all the risks have been mapped onto the risk taxonomy (sometime the list may end up with hundreds of lines), institutions (usually Risk Management teams in conjunction with business teams) are required to provide their own assessment of the **materiality** (also called **criticality**) of each of the risks (e.g., high/medium/low – but other denominations may be adopted).

- Materiality is then assessed in two steps: i) without taking into account the **internal control framework** thereby leading to the **inherent risk**, ii) by taking into account the internal control framework to determine the **residual risk**. In principle, the criticality of the residual risk is less or equal (if the control framework lacks efficiency which is always possible) to that of the inherent risk.
- The internal control framework designates all the teams in charge of the controls both located in business functions called **Level 1 control** (e.g., heads of trading desks are entitled to review – usually on a sample basis- the correct marking of the derivatives book performed by the traders reporting to them) and **non-business functions** (also called **independent control** functions or **Level 2 control**). For instance, the Level 2 control will perform alternative derivative valuation and confront them to those which have been already reviewed by the head of the trading desk and adjust the valuation if needed. For the sake of clarity, there is exist the **Level 3 control** also known as internal audit, or general inspection and perform specific task (e.g., end-to-end review process) and has a direct report to the CEO to ensure full independence. The existence of 3 level of controls is mandatory and framed by the regulations in place.

²⁵ For instance: Risk Control Self-Assessment (“RCSA”) is one popular risk taxonomy adopted by many banks.

The distinction between inherent and residual risk is key: the absence of internal control is synonymous to a maximal risk (i.e., immediate and non-sustainable profit-driven activity) to the brink of mismanagement hence exposing the shareholders to a support a sizeable amount of unexpected losses and thereby over sizing, even spoiling, the capital to operate. The internal control framework aims at ensuring that all business processes remain under control (e.g., loans are granted to creditworthy clients, derivative valuation are appropriate, cyber risks are monitored) to reduce potential unexpected losses and capital allocated by the shareholders.

In other words, let us add this crucial precision: regulatory capital (Pillar 1 & Pillar 2) is designed to absorb unexpected exposures attributable to residual risks and not inherent risks.

The ICAAP requires to include at a bare minimum all **inherent material risks** following the Step 1 of the Figure #25 to ensure completeness of risk assessment. Inherent risks and residual deemed non-material / less material might be disregarded - but this choice is left upon each institution decision and may be jeopardized by the supervisors if the justification to omit a risk is poorly documented. Then the institution is required to determine whether these residual risks (for each risk incorporated in the step #1) have been consistently measured to determine whether the level of Pillar 1 Capital is sufficient by designing new / alternative measurement methods (again under 1 year time horizon and 99.9% confidence level for Credit Risk) or tweaking the existing Pillar 1 formulas. As for those risks absent from Pillar 1, institutions are required to design new measurement techniques. Let us illustrate this stance with some examples to be more concrete:

- Recall that the Credit-VaR presented in Figure #20 is grounded on an assumption of “infinitely grained loan portfolio” (i.e., granular loan portfolio in which there is no concentration / no loan whose size represents a significant amount of the portfolio). In practice, some institutions may be faced with concentration and need to determine (the method remains proprietary for each institution and not exposed here) a higher level of capital than that stemming from Figure #20. The difference between the proprietary method and the Credit VaR corresponds to a capital add-on.
- Similarly, if an institution observes that, say 5% of its total incurred Op Loss ²⁶over an historical set of 5 years exceeds the Pillar 1 Capital attributable to Op Loss, it may consider an add-on equal to the difference between, the 5% quantile of incurred Op Loss and the Pillar 1 capital.

Once all capital add-ons have been determined, they can be added (or sub-added if the institution is willing to account for a diversification effect between risk categories) to come up with the Pillar 2 capital.

For tractability reasons, the Pillar 2 capital add-ons are expressed under percentages added to the solvency ratio as allocated between the CET 1, AT 1, T2– following a dialogue with the supervisors. If you consider Figure #21 again, you observe that there is a footnote indicating the capital add-ons are accounted on top of the each of the solvency “sub-ratios”.

The step #2 requires projecting the solvency ratio over 3 years in accordance with the institutions’ strategic business plan to assess its robustness under **baseline conditions**. The projection is made without any stress and it is expected that the solvency ratio never falls under the **Total Solvency Ratio** (“TS” equal to 8% + capital add-ons + Conservation + Countercyclical buffers), reaching hence a minimum clearly above 8%. In other words, the ICAAP is part of the strategy to ensure that the business plan is sustainable. The leverage ratio (also a capital ratio) shall remain above 3%.

Step#3 is a projection under **stressed conditions** designed by the institution under the assumptions that the stress is “severe but plausible”. In that latter case, the solvency ratio should never go below 8% +

²⁶ Supervisors impose to all institutions to design a framework (and IT system) to declare, measure and report losses attributable to Op Risks; in that regard institution have strong historical dataset regarding their Op Losses.

capital-add-ons. Conservation and / or Countercyclical buffer may be eroded in that occasion. However, the leverage ratio shall remain above 3% under stressed conditions as well.

Finally, let us emphasize again that once the add-ons have been determined by the institution, then the ICAAP document (including all three steps and the add-on computation) is examined and validated (sometimes amended) by the supervisor, then the solvency ratio thresholds become binding (e.g., 4.5% for CET 1 + add-ons, 6% for T1 + add-ons) and Pillar 1 do not hold any longer²⁷.

- For instance: if the outcome of the ICAAP is an add-on of 0.5% on the CET1, the bank should never have a level of CET 1 capital falling below 5% (4.5%+0.5%) of its RWA (otherwise it is no longer allowed to operate and the supervisor withdraws its banking license).

Worth noting that, institutions should update their ICAAP on a quarterly basis – some institutions, less complex, amend the ICAAP on a yearly basis.

3.4.2 Pillar 2 on Liquidity – ILAAP

Pillar 2 on liquidity is conducted in the same vein as for the Capital to come up with a bespoke approach on liquidity. Figure #26 illustrate the 3 steps composing the ILAAP exercise.

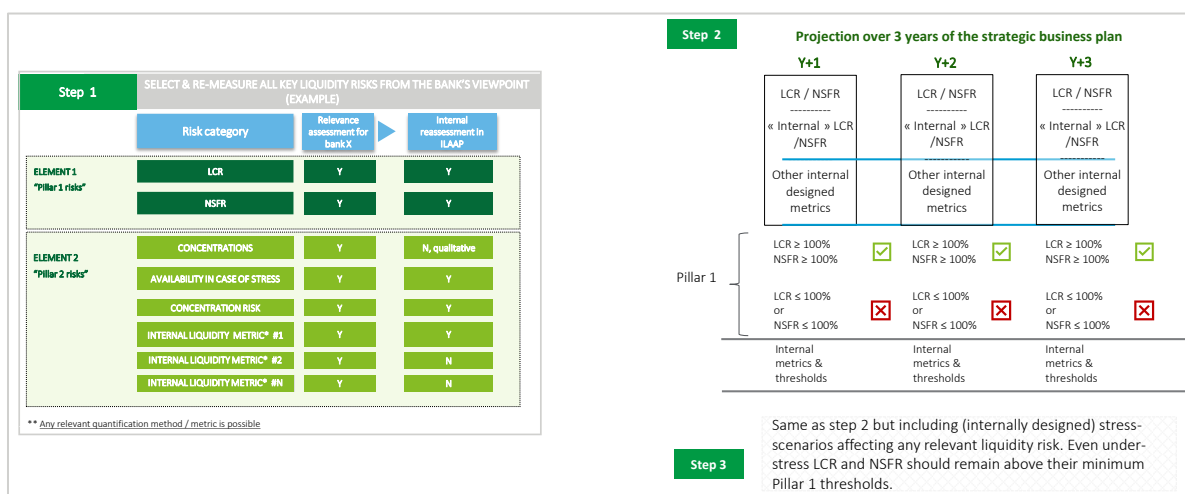


Figure 26 – ILAAP building blocks

ILAAP is also articulated around 3 steps, where Step#1 aims at assessing **all material liquidity risks**, such as the existence of funding concentration and/or the reliability of certain sources of liquidity under periods of stress and how to mitigate those risks.

Institutions are encouraged to tweak / challenge the regulatory coefficients applicable to each liability and asset side component to come up with their “internal” NSFR and LCR. Banks may also incorporate into their ILAAP their own proprietary liquidity metrics used to manage the liquidity risk. In Step #2 and #3, institutions are required to project their NSFR and LCR over a 3-year horizon in accordance with their strategic business plan used for the ICAAP both under a baseline and a stress scenario. This exercise aims at pinpointing and quantifying liquidity risks and design appropriate mitigation measures and determine their funding needs.

In contrast to the ICAAP, in case “internal” NSFR and LCR computations turn out to be **lower** than Pillar 1 (100% for both ratios), supervisors at least in the EU, are not entitled to require binding liquidity add-on to raise regulatory NSFR and LCR above 100% in such a manner that “internal” LCR and NSFR remain above

²⁷ The interested reader may consult the Pillar 2 add-ons for institutions directly supervised by the ECB [Pillar 2 requirement \(europa.eu\)](https://www.ecb.europa.eu/press/pr/2014/09/20140910_pillar2_addons_en.html).

100% as well. Nevertheless, the supervisor must validate the ILAAP content and may, following a dialogue with the institution impose certain measures such as a diversification of the funding sources.

3.5 How do supervisors ensure that the Pillar 1 are ensured in practice?

Regulatory wise, institutions are supposed to permanently comply with their regulatory constraints, i.e., their capital ratios and liquidity ratios should remain above the minimum thresholds. However, how are supervisory bodies (e.g., ACPR, ECB) informed on a regular basis that an institution complies with its regulatory requirements?

There exist a series of **regulatory reportings**, which as opposed to Pillar 3 **disclosures**, are not public information (unless an institution is willing to disclose some of them in their Pillar 3) provide a series of detailed figures on the components and sub-components of their RWA (e.g., detail on the RWA credit by type of client – Retail, Corporate), available capital (e.g., CET1, AT1). Those risk-related figures are transmitted to the supervisory bodies on a quarterly basis in the **Common Report (“COREP”)**. More precisely, the COREP presents the main risks (credit, market and operational), solvency & leverage ratios (capital ratios) and liquidity ratios (LCR and NSFR). The COREP’s framework is considered as the main and most important regulatory report, it serves as a “risk identity card” and ensure that credit institutions are in a capacity to operate.

In parallel, credit institutions also provide a granular view of their financial statements under the **Financial Report (“FINREP”)**. COREP and FINREP are updated and transmitted on a quarterly basis.

Supervisors perform (automated) controls and compare COREP and FINREP across credit institutions. Other reports are also transmitted to the supervisors such as:

- **Reporting Unifié des Banques & Assimilés (“RUBA”)** in force since January 31st, 2022 (and replacing the Système Unifié de Reporting Financier (“SURFI”)) is a French reporting produced solely for the Banque de France (“BdF”) and for the French banking supervisory body (ACPR). This report mixes numerous kind of data: accounting data, statistical data used by the BdF to monitor the monetary and financial activity, non-financial data (e.g. headcounts, business locations).
- **Analytical Credit Datasets (“Anacredit”)** reports to the ECB, for each credit institution’s counterparty (retail clients are excluded) the total exposure composed of Anacredit eligible instruments (derivatives are excluded). Specific information about Counterparties (e.g. nature, ultimate parent company), collaterals & guarantees are provided on a granular level. The ECB uses Anacredit as a tool to drive its monetary policy and macroprudential supervision.
- **Green Asset Ratio (“GAR ”)** is a newly EU established ratio which covers asset and off-balance sheet exposures. This metric’s objective is to make sure that bank’s assets are aligned with a green taxonomy provided by the regulators, in other words it indicates how “green” is the Balance sheet.

You may here consider that the data quality increasing relies at the heart of the banking business and the supervisors scrutinize all progresses made by the institutions in that regard.

3.5 Beyond Pillar 1 & Pillar 2: recovery and resolution framework

No development this year as not addressed during the lecture.

3.6 Banking regulation – the main actors

The idea of the section is to provide a simple overview related to the legislative and executive mechanisms underpinning the banking regulation. It does not aim to expose in great length the features of the regulation but rather focus on the main actors which are: i) the Basel Committee, ii) the EU, iii) the US.

3.6.1 Basel Committee

The main message from this section is that the Basel Committee sets out a series of principles framing the banking regulation. However, these principles are not enforceable until they are transformed into a legal framework adopted by the countries / geographies willing to adopt the Basel Rules.

3.6.2 Banking Regulation & Supervision in the EU (European Union)

Let us come back to the beginning of section 2 whereby we briefly introduced the articulation between the Basel Accords and their translation into the European Law under a series of regulation and directives (Capital Requirement Regulation and Directive).

The Banking Union established in 2012 is made of 3 vertices (building blocks) presented in the Figure #27.

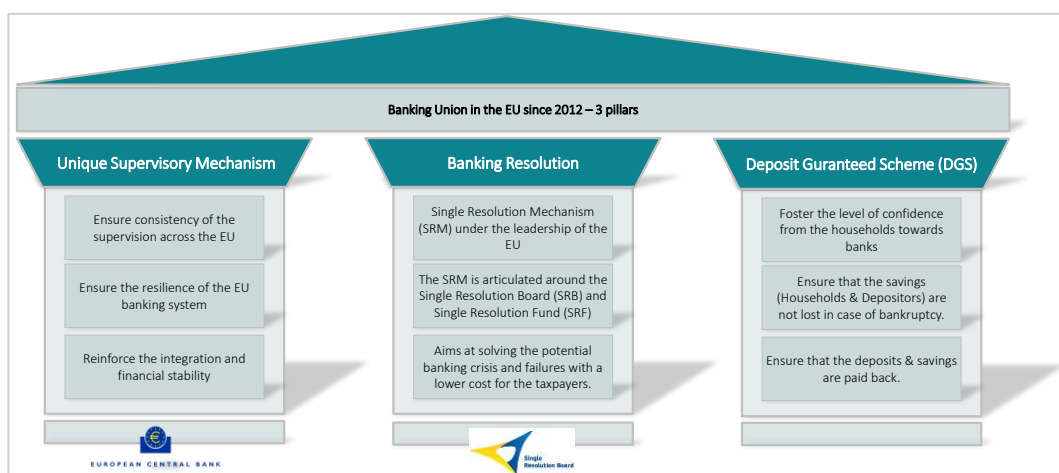


Figure 27 – Banking Union in the EU

Let us focus on the European Central bank (“ECB”) who is the ultimate **supervisor** who ensures that all banks operating in the EU comply with the laws and regulation (e.g., CRR) proposed by the **regulator**. A **supervisor** is hence granted by **executive powers** whilst a **regulator** is bestowed with **legislative powers**. In France, the ACPR is the supervisor but it has been placed since 2014 under the umbrella of the ECB. Some supervisory powers remain in the hands of the ACPR and some are directly endorsed by the ECB. In practice, supervisor and regulator may be used without distinction but we recommend to use these terms in accordance with their correct meaning.

In the EU the European Commission, Council and Parliament (collectively denoted “Trilogue” when it comes to banking regulation) set the tone and are empowered to translate the Basel Accords into the CRR and CRD. Recall that EU Regulation are applicable by all EU members once voted by the European Parliament, whilst CRD should be translated (and sometimes amended to a limited extent) by each EU member into its own legal regime and voted by its parliament.

Besides, the European Banking Authority (“EBA”), was created by the European Parliament and the Council of the EU in 2010 (and is headquartered in Paris). The EBA is responsible for promoting supervisory convergence and enhancing the integrity, transparency, and consistency of banking supervision across the EU.

In that regard, the European Commission delegates (and this is written as such in the CRR) to the EBA its powers to setting supervisory standards and guidelines, conducting supervisory reviews and assessments, and providing technical advice to EU institutions and **national competent authorities** (NCAs, e.g. the ACPR in France). The EBA also publishes binding regulatory technical standards (“RTS”)

which bring additional precisions (and sometimes mathematical formulas) to determine certain type of risks when computing a capital and / or a liquidity ratio. As for the ICAAP/ILAAP, the EBA issues Guidelines to be followed by the institutions. The **EBA is hence acting as a regulator** even if his role is more blurred. Please refer to the appendix for more precisions around the delineation from Basel Agreements to EBA.

As for the two other vertices, the **Single Resolution Board (“SRB”)**, independent from the ECB, prevents banking bankruptcy through so-called “**resolution powers**” to take the helm of any ailing institution (an expel its Senior Management), i.e. in a “near bankrupt” state and is entitled to undertake all necessary efforts to not liquidate the bank. In that regard, the SRB is endowed with a Resolution Fund to exert its powers and may use some resolution strategies such as the “bail-in” which consists in converting certain debts into Capital to restore solvency ratios. We will not delve into the SRB and resolution powers in this document.

The **Deposit Guaranteed Scheme (“DGS”)** aims at protecting retail depositors (up to a certain threshold, e.g. EUR 100,000 in France) in case a bank would default.

3.6 Banking Supervision & Supervision in the US

In the US, the Congress plays a significant role in the endorsement and implementation of the Basel agreements. Whilst the Basel agreements are not legally binding in the US, the Congress has the authority to adopt legislation that incorporates these agreements into **federal law** and gives **regulatory agencies** the authority to implement and enforce them.

For example, the Dodd-Frank Wall Street Reform and Consumer Protection Act, which was signed into law in 2010, included **provisions that required the federal bank regulatory agencies to adopt regulations implementing the Basel III capital and liquidity standards in the United States**. These regulations, which were issued by the Office of the Comptroller of the Currency (OCC), the Federal Reserve System (FED), and the Federal Deposit Insurance Corporation (FDIC), set out the specific requirements that banks must follow in order to comply with the Basel III capital and liquidity standards.

In addition to adopting legislation that incorporates the Basel agreements into federal law, the Congress also has oversight authority over the regulatory agencies that are responsible for implementing these agreements.

More precisely, the main agencies which both endorse a role of supervisor and regulator are as follows (their perimeter depends on the **typology of financial institution** – as you may note the typology is broader than that in place in the EU, thereby reflecting the essence of a federal country which is the US - placed under their scope).

- Office of the Comptroller of the Currency (OCC): The OCC is a federal agency that is responsible for chartering, regulating, and supervising **national banks and federal savings associations**.
- The Federal Reserve System (FED): The Fed is the central bank of the United States and is responsible for implementing monetary policy, regulating and supervising banks, and providing financial services to the government.
- The Federal Deposit Insurance Corporation (FDIC): The FDIC is an independent agency that is responsible for **insuring deposits at banks and savings institutions**, examining and supervising financial institutions for safety and soundness, and implementing and enforcing consumer protection laws.
- The National Credit Union Administration (NCUA): The NCUA is a federal agency that is responsible for chartering, regulating, and **supervising federal credit unions**.

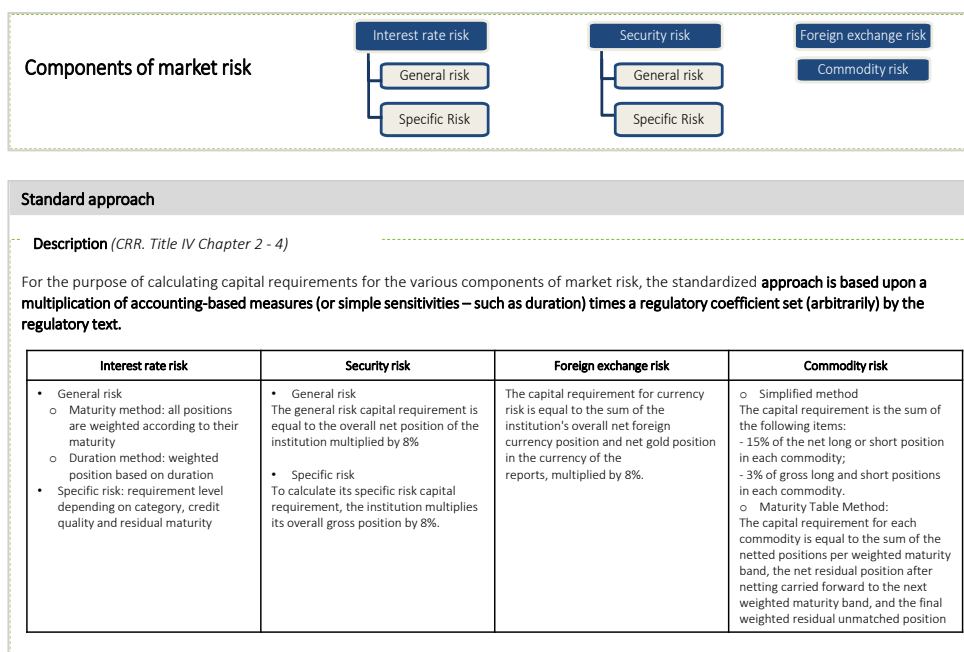
4 Conclusion

You may have observed that realm of banking regulation is a complex subject necessitating a broad range of skills (business, regulation, mathematics and above all data, data and data) to ensure a comprehensive view on the subject. You are at the start of your journey, and I can only encourage you to have a look at the appendix and / or the bibliography in case you are willing to go more in depth. Needless to mention that the next lecture will come back to some notions presented in this document and will be more focused on the mathematical techniques to measure risk.

5 Appendix

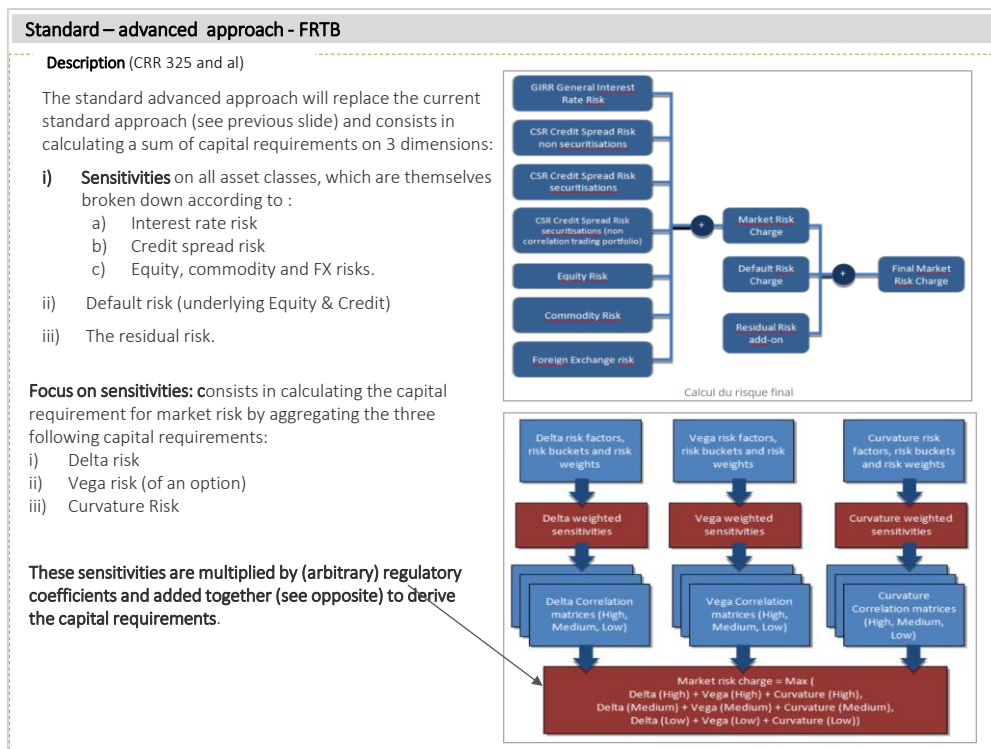
5.1 Additional details on Market Risk (standard approaches) on Pillar 1

The figure below provides an outline underpinning the **current standard market risk approach**. We do not provide with the computation details but solely the risks pinpointed by the CRR.



Recall that derivative trading business do not lie at the heart of credit institutions, and nothing forces them to engage in such activities. Therefore, those credit institutions engaging into trading activities do it on purpose and adopt the internal approach based on the VaR and barely use the standard market risk approach.

The figure below provides an outline underpinning the **upcoming new market risk approach (Fundamental Reform of the Trading Book ("FRTB") standard**. Again, we do not provide with the computation details but solely with the risks pinpointed by the CRR



5.2 Additional details to measure Counterparty Credit Risk (“CCR”) on Pillar 1

The figure below provides some details prevailing to compute the EAD when it comes to derivatives exposures.

Institutions shall calculate the credit risk stemming from exposure value of derivatives contracts using one of the methods set out thereafter. Recall that, as opposed to Loans, derivative value vary overtime in accordance with market conditions (and even pivot from negative to positive and vice versa). Henceforth, measuring their exposure EAD is difficult.

4 methods are available: i) two **Standardized methods** (“SA-CCR”), ii) the **Original Exposure Method** (“OEM”) aim at calculating the exposure value (EAD) in some specific prescribed ways without requiring any simulation and iii) the **Internal Model Method** (“IMM”) which requires to compute the evolution of the mark-to-market of a portfolio of derivatives over a 1-year time horizon to derive the EAD.

Those SA-CCR and OEM are jointly denoted as “non-IMM”.

The 3 “non-IMM” rely upon the same starting formula:

$$EAD = \alpha * (Replacement\ cost\ (RC) + Potential\ Future\ Exposure(PFE))$$

Recall that once the EAD has been determined, it multiplied by the corresponding regulatory RW (standardized approach for credit risk) or input into the Credit VaR formula to come up with the corresponding capital requirements.

Let us elaborate a little more on the various approaches.

OEM

- Conditions of application: the total balance sheet and off-balance sheet of derivatives activities must be simultaneously: ≤ 100 million euros and represent less than 5% of the institution's total assets;

- The exposure value (EAD) is calculated separately for each **netting set** as follows for the Replacement Cost (“RC”):
 - $RC = TH + MTA$ for **Listed and OTC derivatives** where: i) TH corresponds to the minimum threshold below which the margin call is not authorized, ii) MTA: Minimum Transfer Amount.
 - $RC = \text{Max}(CMV, 0)$ for other derivatives, where CMV is the current market value of the derivative.

A **netting set** allows the institution to algebraically add the positive and negative derivatives values to reduce the final exposure, per counterparty. A contractual agreement called “netting agreement” should be in place to perform such a computation; we will not elaborate further on the contractual aspects.

The Potential Future Exposure (“PFE”) is equal to the **notional amount** of the derivative, multiplied by a fixed coefficient that depends on the asset class and the residual maturity.

- Ex: for an interest rate swap: $0.5\% \times \text{Residual maturity}$; 32% for equity derivatives.

2 SA-CCR methods (regular and simplified)

The approach is based on the Market Value of the derivatives, the margin calls, their asset class, notional and does not require performing any simulation.

Conditions for the application of the **simplified SA-CCR** approach: the total balance sheet and off-balance sheet of derivative activities must be simultaneously: ≤ 300 million euros and $\leq 10\%$ of the institution's total assets. If those conditions are not satisfied the institution mandatorily applies the regular SA-CCR.

Again, we will not present these methodologies which may appear as a conundrum to compute both the RC and PFE at first glance, but we encourage the interested reader to consult [4] (articles 274 to 283).

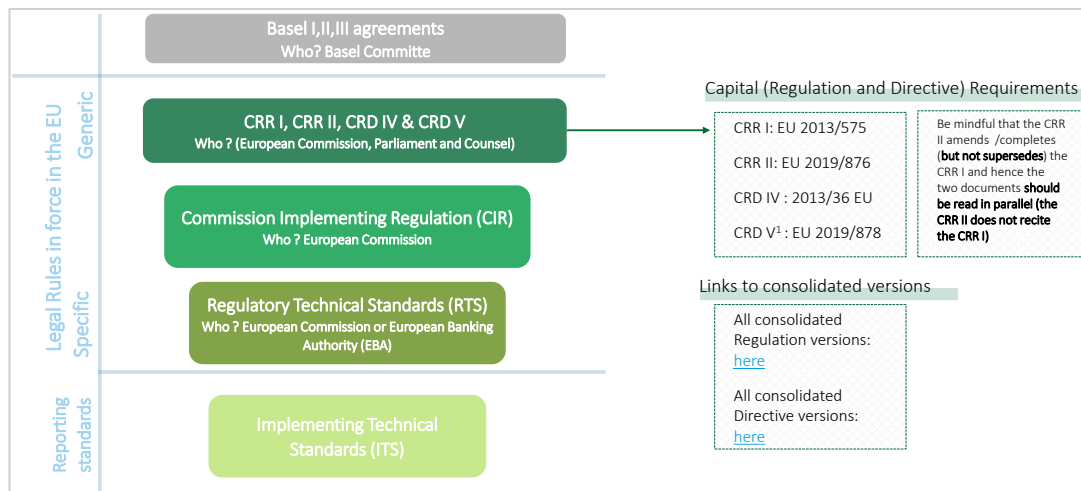
IMM

The approach is based on an internal model (simulation of the of the positive derivatives values - called « Expected positive exposures ») including margin calls of the bank which must meet quantitative, qualitative and data requirements. As for the IRB for Credit Risk, the IMM requires the validation from the supervisor (e.g. ACPR or ECB).

5.3 From Basel Agreements to EBA

The figure below illustrates how the Basel Regulations are delineated and enforced in the EU. In a nutshell, the Basel Agreements are translated into EU legal framework (Regulation, Directives and their translation into each EU members’ national law). Following a certain number of rounds of discussion between the EU Commission, Council and Parliament (the “trilogue”), the final Regulation(s) / Directive(s) are voted by the EU Parliament to enter into force.

Usually, the CRR / CRD require the European Banking Authority (“EBA”) to produce additional technical content directly endorsable by the EU Commission (without being subject to a vote from the EU Parliament). This technical content is articulated around Regulatory Technical Standards (“RTS”) and Implementing Technical Standards (“ITS”) which define in great lengths the regulatory reporting details (e.g. COREP, FINREP).



6 Bibliography

6.1 Textbooks

[1] Poncet, P, Portait, R (2023) Capital Market Finance, Springer

[1.1] Poncet, P, Portait, R (2023), Finance de Marché – 5^{ème} édition, Dalloz

[2] Gregory, J., (2020) The xVA Challenge, Wiley

6.2 Some key regulatory texts

6.2.1 EU regulation on solvency & liquidity

[3] EU Regulation 575/2013 on Prudential on prudential requirements for credit institutions and investment firm (**Capital Requirement Regulation (“CRR I”)**)

[4] EU 2019/876 amending Regulation (EU) No 575/2013 (...) as regards the leverage ratio, the net stable funding ratio, requirements for own funds and eligible liabilities, counterparty credit risk, market risk, exposures to central counterparties, exposures to collective investment undertakings, large exposures, reporting and disclosure requirements (**Capital Requirement Regulation (“CRR II”)**)

[5] EU 2024/1623 amending Regulation (EU) No 575/2013 (...) as regards requirements for credit risk, credit valuation adjustment risk, operational risk, market risk and the output floor (**Capital Requirement Regulation (“CRR III”)**)

[6] EU Directive 2013/36/EU on access to the activity of credit institutions and the prudential supervision of credit institutions and investment firms (**Capital Requirement Directive (“CRD IV”)**)

[7] EU Directive 2019/878 amending Directive 2013/36/EU as regards exempted entities, financial holding companies, mixed financial holding companies, remuneration, supervisory measures and powers and capital conservation measures (**Capital Requirement Directive (“CRD V”)**)

[8] EU Directive 2024/1619 amending Directive 2013/36/EU as regards supervisory powers, sanctions, third-country branches, and environmental, social and governance risks (**Capital Requirement Directive (“CRD VI”)**)

Note : interactive version of all these documents are available here: [Interactive Single Rulebook | European Banking Authority \(europa.eu\)](#)

6.2.2. EU Guidelines related to ICAAP & ILAAP

[9] ECB Guide to the internal capital adequacy assessment process (ICAAP), November 2018

[10] ECB Guide to the internal liquidity adequacy assessment process (ICAAP), November 2018

[11] EBA/GL/2016/10 - Guidelines on ICAAP and ILAAP information collected for SREP purposes

6.3 Websites

Basel Committee: www.bis.org (>> Committees & Associations >> Basel Committee for Banking Supervision (BCBS) >> Basel 3)

European Banking Authority (EBA): www.eba.europa.eu

7 Index

B

bank. See Credit Institution
Basel

CRR, CRD, 4, 18, 20, 21, 28,
35, 36, 37, 40, 42

C

Capital

Equity, Regulatory Capital, 4,
7, 18, 20, 21, 22, 23, 26, 27,
28, 32, 33, 34, 36, 37, 41

Counterparty Credit Risk

Counterparty Risk, 26, 39

Credit Institution, 6, 13, 21, 35

Credit Risk, 8, 11, 12, 13, 18, 26,
39, 41

Credit VaR, 24, 25, 33, 39

E

Economic Capital

ICAAP, Pillar 2, 18, 20, 31

Expected Loss, 9, 23

Expected Shortfall, 26

F

Fundamental Reform of the
Trading Book
FRTB, 26, 38

I

ILAAP, 21, 34, 35, 37, 42

institution. See Credit
Institution

L

Leverage Ratio, 28, 29

Liquidity Coverage Ratio

LCR, 20, 30

Liquidity Risk

Funding Risk, 16, 17

Loss Given Default

LGD, Recovery Rate, 23

M

Market Risk, 12, 13, 15, 22, 23,
25, 26, 38

Market VaR, 25, 26

O

Operational Risk

Op Risk, 22, 23, 25

P

Pillar 1, 21, 22, 28, 29, 31, 32, 33,
34, 35, 38, 39

Pillar 2, 21, 28, 30, 31, 32, 33, 34,
35

R

Risk Weighted Assets

RWA, 22

Risk Weights

RW, 24

S

Solvency Ratio, 22, 33

V

Value at Risk

VaR, 24